

# BDAT 614: Statistical Quality Control and Improvement

## Statistical Methods for Process Monitoring, Control & Excellence

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June 5, 2026

# Course Description & Relevance

- Comprehensive introduction to quality control using statistical methods
- Key topics: Process variability, Control Charts, Process Capability, Acceptance Sampling, Six Sigma & Lean, TQM
- Big Data Focus: Real-time SPC, IoT/sensor data, Spark streaming, ML-enhanced monitoring
- Goal: Equip students with industry-ready skills for data-driven quality improvement

# Learning Objectives

- Present fundamental principles of Statistical Process Control (SPC)
- Design, construct and interpret control charts for variable and attribute data
- Evaluate process capability and performance
- Apply statistical techniques to identify and resolve quality issues
- Integrate classical SQC with modern Big Data Analytics tools

# Course Overview

- Fundamental concepts of Quality, Quality Control, and Quality Assurance
- Dimensions of Quality and their importance in data-driven environments
- Introduction to Statistical Quality Control (SQC)
- Introduction to Quality and Variation
- Statistical Process Control (SPC) Fundamentals
- Control Charts: Detailed Design & Examples
- Process Capability Analysis

# Course Overview

- Shewhart Cycle (PDCA/PDSA)
- DMAIC Framework
- Measurement System Analysis (Gauge R&R with ANOVA)
- Acceptance Sampling
- Six Sigma, Lean Tools & DMAIC
- Continuous Improvement and Total Quality Management (TQM)
- Big Data Applications in SQC
- Summary and Case Studies

# What is Quality?

## Classic Definitions

- **Fitness for Use** — Joseph Juran
- **Conformance to Requirements** — Philip Crosby
- **Delight the Customer** — Modern perspective
- **Predictable Degree of Uniformity and Dependability** — W. Edwards Deming

**In Big Data Context:** Quality of data, algorithms, models, and insights.

# Dimensions of Quality (David Garvin)

- 1 **Performance** — Primary operating characteristics
- 2 **Features** — Secondary characteristics
- 3 **Reliability** — Probability of failure-free operation
- 4 **Conformance** — Meeting specifications
- 5 **Durability** — Useful life
- 6 **Serviceability** — Speed and ease of repair
- 7 **Aesthetics** — Look, feel, sound, taste, smell
- 8 **Perceived Quality** — Reputation and image

**Note:** Different stakeholders may prioritize different dimensions.

# Quality Control vs Quality Assurance

## Quality Control (QC)

- Reactive
- Product-oriented
- Detection of defects
- Statistical tools, inspection, testing
- "Find and fix"

## Statistical Quality Control (SQC)

- Uses statistics to monitor and control processes

## Quality Assurance (QA)

- Proactive
- Process-oriented
- Prevention of defects
- System design, audits, standards (ISO 9001)
- "Build quality in"

# ISO 9001: Quality Management System

**ISO 9001** is an internationally recognized standard for **Quality Management Systems (QMS)** developed by the International Organization for Standardization (ISO).

## Current Version:

- ISO 9001:2015

## Purpose of ISO 9001

- Ensure consistent product and service quality
- Enhance customer satisfaction
- Improve operational efficiency
- Promote continual improvement
- Meet regulatory and statutory requirements

# ISO 9001: Quality Management System

## Key Principles

- Customer focus
- Leadership
- Engagement of people
- Process approach
- Continuous improvement
- Evidence-based decision making
- Relationship management

## Benefits of Certification

- Increased credibility and competitiveness
- Improved risk management
- Better process consistency
- Enhanced organizational performance

# Introduction to Statistical Quality Control (SQC)

- Application of statistical methods to monitor, control, and improve quality
- Pioneered by Walter A. Shewhart (1920s)
- Key Contributors: Shewhart, Deming, Juran, Ishikawa
- Core Philosophy: **Reduce Variation**

**Goal:** Make processes predictable and capable of meeting customer requirements.

# Process Variation – The Heart of Quality Control

*"Variation is the enemy of quality." — W. Edwards Deming*

All processes exhibit variation. Understanding, measuring, and reducing variation is the foundation of Statistical Quality Control.

# What is Process Variation?

- Natural differences in process output even when inputs are the same
- No two products or services are ever exactly identical
- Measured as dispersion (spread) around the target value
- Sources: Raw materials, machines, methods, people, environment, measurement systems

**Goal of SQC:** Reduce variation to improve consistency and predictability.

# Two Types of Process Variation

## Common Cause Variation

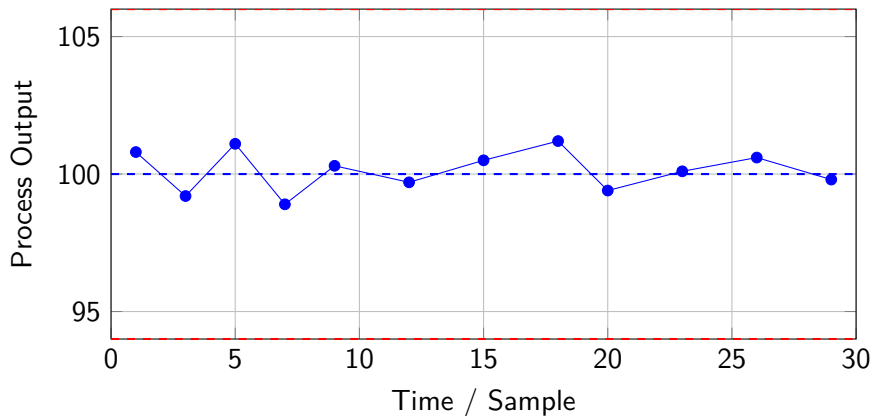
- Inherent in the process
- Random and predictable
- Many small, unavoidable causes
- Process is **stable** (in statistical control)
- Requires system-level improvement

## Special Cause Variation

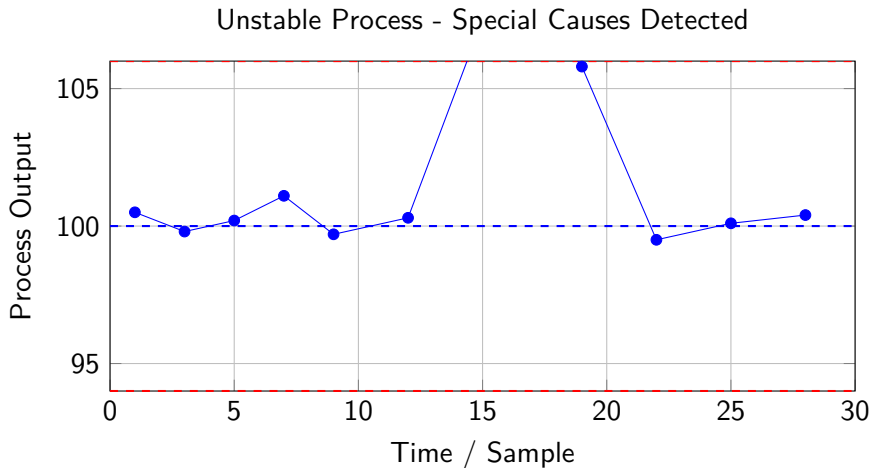
- Assignable, specific causes
- Non-random, unusual events
- Detectable on control charts
- Process is **out of control**
- Can (and should) be eliminated

# Visualizing Process Variation

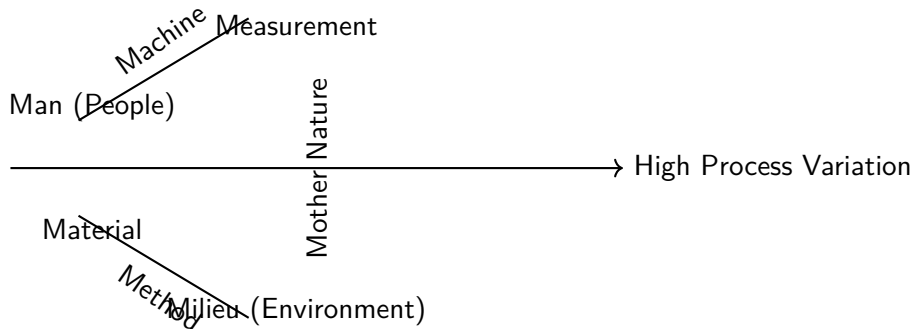
Stable Process - Only Common Causes



# Unstable Process – Special Causes Present



# Sources of Variation – The 6Ms (Fishbone)



# Impact of Variation on Quality

- Increases defects and rework
- Reduces process capability ( $C_p$  and  $C_{pk}$ )
- Leads to customer dissatisfaction
- Increases costs (Cost of Poor Quality)
- Makes prediction and planning difficult
- Hides root causes of problems

**Rule of Thumb:** Reducing variation is often more important than centering the process.

# Strategies to Reduce Variation

- ① Identify and remove Special Causes (using Control Charts)
- ② Reduce Common Cause Variation through:
  - ▶ Process redesign and standardization
  - ▶ Better training and methods
  - ▶ Improved equipment maintenance
  - ▶ Supplier quality improvement
  - ▶ Environmental control
- ③ Apply DMAIC / Six Sigma projects
- ④ Use Big Data for deeper variation analysis

**Note:** You can always use the 5 Whys technique to dig to the root cause of all problems.

# The 5 Whys Technique

- A simple yet powerful root cause analysis tool
- Developed by Taiichi Ohno at Toyota as part of the Toyota Production System (TPS)
- Core Idea: Ask "**Why?**" repeatedly (typically 5 times) to drill down from symptoms to the fundamental/root cause
- Helps move from treating **symptoms** to solving the real **root cause**.

# Purpose of 5 Whys in Manufacturing

- Identify the true root cause of defects, downtime, quality issues, or safety incidents
- Prevent recurrence of problems
- Support continuous improvement
- Simple, no statistical software required and can be used on the shop floor
- Complements advanced tools like Fishbone Diagram, Pareto Analysis, and DMAIC

# Step-by-Step Process of 5 Whys

- 1 Clearly define the **problem** (be specific)
- 2 Ask "**Why?**" — record the answer
- 3 Ask "**Why?**" again for the previous answer
- 4 Repeat until you reach the **root cause** (usually 5 times, sometimes 3–7)
- 5 Develop **corrective and preventive actions**
- 6 Assign **responsibility** (tagging) and timeline for resolution
- 7 Verify effectiveness after implementation

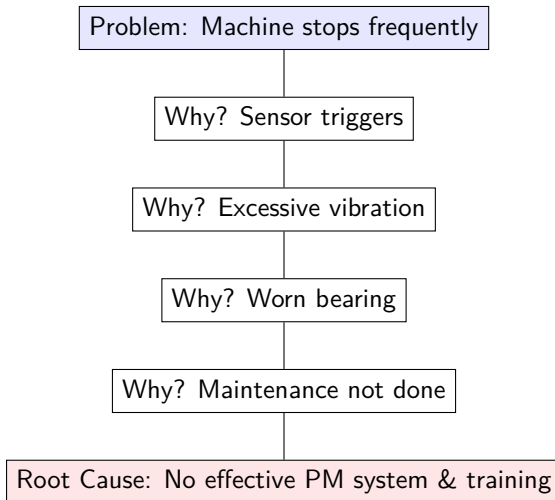
## 5 Whys – Manufacturing Example

**Problem:** Machine keeps stopping unexpectedly during production.

- Why 1: The machine overload sensor is triggering.
- Why 2: There is too much vibration on the machine.
- Why 3: The bearing is worn out and causing imbalance.
- Why 4: The preventive maintenance schedule was not followed.
- Why 5: There is no proper maintenance tracking system and operators are not trained on importance of scheduled maintenance.

**Root Cause:** Lack of effective preventive maintenance system and training.

## 5 Whys Diagram (Visual)



# Tagging & Resolution (Action Plan)

After identifying the root cause:

- **Tag** responsible person/department
- Define specific, measurable corrective actions
- Set deadlines and verification method

# Best Practices for Effective 5 Whys

- Involve people who actually work on the process (operators, technicians)
- Use data and facts — avoid assumptions
- Keep asking "Why?" until the cause is actionable and controllable
- Do not stop at human error — go deeper (system/process failure)
- Combine with Fishbone Diagram for complex problems
- Document and share learnings across the organization

# Advantages and Limitations

## Advantages

- Simple and easy to learn
- Encourages deep thinking
- Low cost, no special tools needed
- Promotes team collaboration
- Fast root cause identification

## Limitations

- Can be subjective
- May stop too early (not reaching real root cause)
- Not suitable for very complex problems alone
- Requires experience and domain knowledge

## 5 Whys in Big Data & Industry

- Use real-time sensor data to identify problems faster
- Combine 5 Whys with machine learning anomaly detection
- Integrate with DMAIC (Analyze phase)
- Digital 5 Whys tools linked with Computerized Maintenance Management System (CMMS) and ERP systems
- Root cause analysis on large-scale production data

# Key Takeaways – 5 Whys

- 5 Whys is a powerful, simple tool for root cause analysis in manufacturing
- Focus on **system** issues rather than blaming people
- Always follow up with corrective actions and tagging of responsibility
- Effective when combined with other quality tools
- Essential skill for quality professionals and data analysts working in manufacturing

# Variation in Big Data Context

- Sensor drift and calibration issues
- Data pipeline variability (ETL fluctuations)
- Model drift in machine learning systems
- Network latency and cloud resource variation
- Customer behavior variation in service processes

**Opportunity:** Big Data allows real-time detection and much faster reaction to variation.

# Key Takeaways – Process Variation

- All processes have variation and the key is to understand its type
- Distinguish between Common Cause and Special Cause variation
- Control Charts are the primary tool for detecting special causes
- Reducing variation leads to better quality, lower cost, and higher customer satisfaction
- In the Big Data era, variation analysis must be real-time and scalable

# Statistical Process Control (SPC) – Core Concepts

- SPC is a statistical method used to monitor, control, and improve processes
- Developed by Walter A. Shewhart in the 1920s
- Uses **control charts** to distinguish between common and special cause variation
- Foundation of modern quality management (Six Sigma, Lean, TQM)
- Key Principle: "Prevention over Detection"
- 7 Basic Quality Tools: Pareto, Fishbone, Histogram, Check Sheet, Scatter Plot, Flowchart, Control Charts

# Objectives of Statistical Process Control

- Determine whether a process is in a state of statistical control
- Detect the presence of special causes of variation
- Reduce process variability over time
- Improve process capability and predictability
- Provide a basis for continuous improvement (PDCA / DMAIC)
- Support data-driven decision making

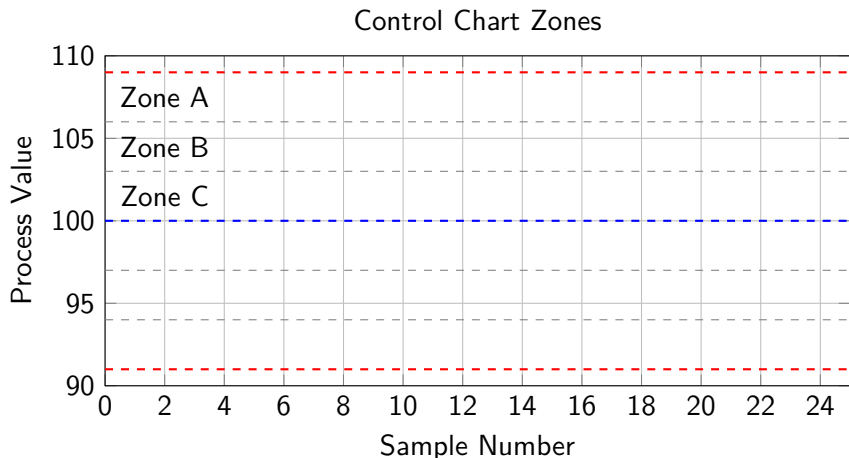
# Benefits of SPC

- Early detection of process problems
- Reduced defects and rework
- Lower operating costs
- Improved product consistency and customer satisfaction
- Better process understanding and knowledge
- Foundation for predictive quality in Big Data environments

# Control Charts – The Heart of SPC

- Graphical tool that plots process data over time
- Shows whether a process is stable or unstable
- Consists of:
  - ▶ Center Line (CL) — usually the process mean
  - ▶ Upper Control Limit (UCL)
  - ▶ Lower Control Limit (LCL)
- Control limits are typically set at  $\pm 3\sigma$  from the centerline

# Control Chart Zones and Interpretation



# Out-of-Control Signals (Western Electric / Nelson Rules)

- 1 Any single point outside the  $\pm 3\sigma$  control limits
- 2 2 out of 3 consecutive points in Zone A or beyond
- 3 4 out of 5 consecutive points in Zone B or beyond
- 4 8 consecutive points on one side of the centerline
- 5 6 consecutive points steadily increasing or decreasing (trend)
- 6 14 consecutive points alternating up and down
- 7 15 consecutive points in Zone C (stratification)

# Control Charts for Variables Data

Variables data = Continuous measurements (weight, length, temperature, time, etc.)

## Most Common Charts:

- $\bar{X}$  and R Chart (subgroup size 2–5)
- $\bar{X}$  and S Chart (larger subgroups)
- Individuals & Moving Range (I-MR) Chart ( $n=1$ )

# $\bar{X}$ and R Chart Example

**Process:** Bottle filling (Target = 500 ml, n=5)

Formulas:

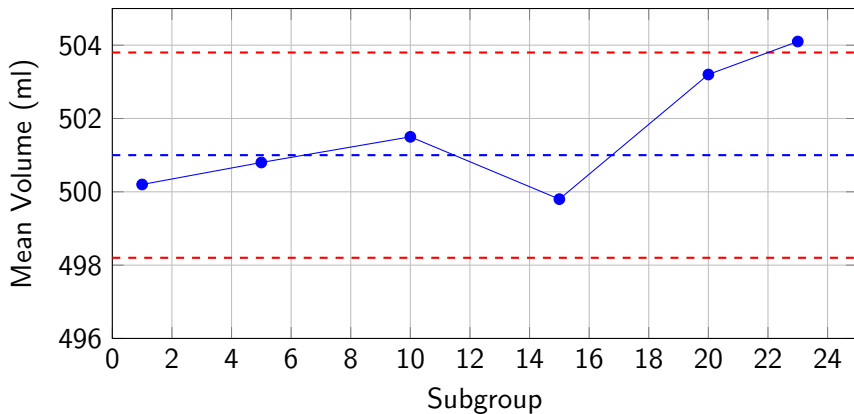
$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$$

$$UCL_R = D_4\bar{R}, \quad LCL_R = D_3\bar{R}$$

( $A_2$ ,  $D_3$ ,  $D_4$  are constants from control chart tables)

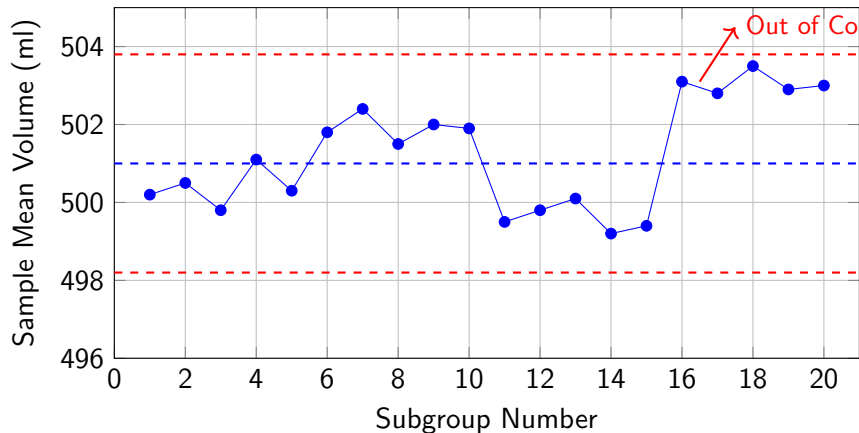
# $\bar{X}$ Chart Visualization (Bottle Filling)

$\bar{X}$  Control Chart

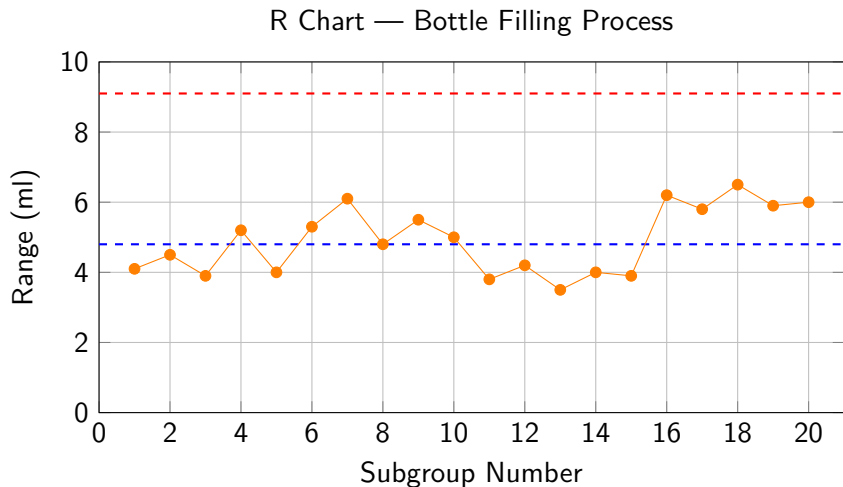


# $\bar{X}$ Control Chart — Visualization

$\bar{X}$  Chart — Bottle Filling Process



# R Chart (Range)



# $\bar{X}$ and R Control Charts

- Most common charts for **variables data** when subgroup size is small ( $n = 2$  to  $5$ )
- $\bar{X}$  Chart: Monitors process **mean** (location)
- R Chart: Monitors process **variability** (range)
- Used together to assess process stability

# $\bar{X}$ and R Chart – Formulas

Let:

- $k$  = number of subgroups
- $n$  = subgroup size
- $\bar{X}_i$  = mean of  $i^{th}$  subgroup
- $R_i$  = range of  $i^{th}$  subgroup

$$\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}, \quad \bar{R} = \frac{\sum R_i}{k}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R}$$

$$UCL_R = D_4 \bar{R}, \quad LCL_R = D_3 \bar{R}$$

## Example: Bottle Filling Process ( $\bar{X}$ -R Chart)

**Process:** Filling bottles with 500 ml target volume. Subgroup size  $n = 5$ ,  
Number of subgroups  $k = 12$

| Subgroup | $\bar{X}$ (ml) | R (ml) |
|----------|----------------|--------|
| 1        | 501.2          | 4.1    |
| 2        | 498.8          | 3.5    |
| 3        | 500.5          | 5.2    |
| 4        | 502.1          | 6.8    |
| 5        | 499.7          | 3.9    |
| 6        | 501.8          | 4.7    |
| 7        | 500.3          | 2.8    |
| 8        | 503.4          | 7.1    |
| 9        | 499.5          | 4.3    |
| 10       | 501.0          | 5.0    |
| 11       | 500.8          | 3.6    |
| 12       | 502.6          | 6.2    |

# Step-by-Step Calculation

$$\bar{\bar{X}} = 500.98 \text{ ml}, \quad \bar{\bar{R}} = 4.77 \text{ ml}$$

For  $n = 5$ :  $A_2 = 0.577$ ,  $D_3 = 0$ ,  $D_4 = 2.114$

$$UCL_{\bar{X}} = 500.98 + 0.577 \times 4.77 \approx 503.73$$

$$LCL_{\bar{X}} = 500.98 - 0.577 \times 4.77 \approx 498.23$$

$$UCL_R = 2.114 \times 4.77 \approx 10.08, \quad LCL_R = 0$$

# Standard Control Chart Constants for Variables Data

| Subgroup<br>Size (n) | $\bar{X}$ -R Chart |                |                | $\bar{X}$ -S Chart |                |                |
|----------------------|--------------------|----------------|----------------|--------------------|----------------|----------------|
|                      | A <sub>2</sub>     | D <sub>3</sub> | D <sub>4</sub> | A <sub>3</sub>     | B <sub>3</sub> | B <sub>4</sub> |
| 2                    | 1.880              | 0              | 3.267          | 2.659              | 0              | 3.267          |
| 3                    | 1.023              | 0              | 2.574          | 1.954              | 0              | 2.568          |
| 4                    | 0.729              | 0              | 2.282          | 1.628              | 0              | 2.266          |
| 5                    | 0.577              | 0              | 2.114          | 1.427              | 0              | 2.089          |
| 6                    | 0.483              | 0              | 2.004          | 1.287              | 0.030          | 1.970          |
| 7                    | 0.419              | 0.076          | 1.924          | 1.182              | 0.118          | 1.882          |
| 8                    | 0.373              | 0.136          | 1.864          | 1.099              | 0.185          | 1.815          |
| 9                    | 0.337              | 0.184          | 1.816          | 1.032              | 0.239          | 1.761          |
| 10                   | 0.308              | 0.223          | 1.777          | 0.975              | 0.284          | 1.716          |
| 15                   | 0.223              | 0.348          | 1.652          | 0.789              | 0.428          | 1.572          |
| 20                   | 0.180              | 0.414          | 1.586          | 0.680              | 0.510          | 1.490          |

Most commonly used values highlighted for  $n = 4, 5, 6$

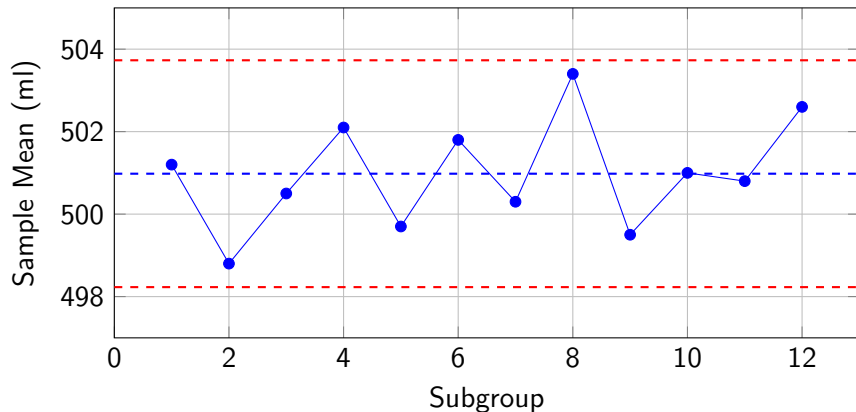
# Control Chart Constants Table (Variables)

| $n$       | $\bar{X} + R$ Chart |       |              | $\bar{X} + S$ Chart |       |              |
|-----------|---------------------|-------|--------------|---------------------|-------|--------------|
|           | $A_2$               | $D_3$ | $D_4$        | $A_3$               | $B_3$ | $B_4$        |
| <b>2</b>  | 1.880               | 0     | 3.267        | 2.659               | 0     | 3.267        |
| <b>3</b>  | 1.023               | 0     | 2.574        | 1.954               | 0     | 2.568        |
| <b>4</b>  | 0.729               | 0     | 2.282        | 1.628               | 0     | 2.266        |
| <b>5</b>  | <b>0.577</b>        | 0     | <b>2.114</b> | <b>1.427</b>        | 0     | <b>2.089</b> |
| <b>6</b>  | 0.483               | 0     | 2.004        | 1.287               | 0.030 | 1.970        |
| 7         | 0.419               | 0.076 | 1.924        | 1.182               | 0.118 | 1.882        |
| 8         | 0.373               | 0.136 | 1.864        | 1.099               | 0.185 | 1.815        |
| <b>10</b> | 0.308               | 0.223 | 1.777        | 0.975               | 0.284 | 1.716        |

Source: Standard control chart constant tables (ASTM, Montgomery)

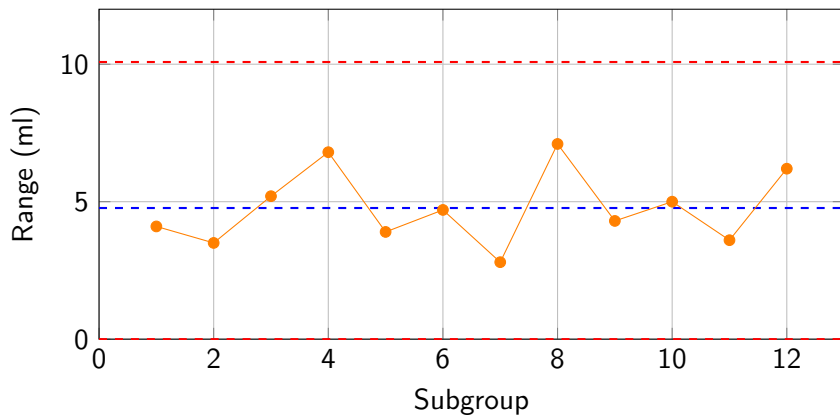
# $\bar{X}$ Control Chart

$\bar{X}$  Chart — Bottle Filling Process



# R Chart

R Chart — Bottle Filling Process



# $\bar{X}$ and S Control Charts

- Preferred when subgroup size  $n > 10$
- S Chart uses **standard deviation** instead of range → more efficient for larger samples
- Better estimate of process variability

## $\bar{X}$ and S Chart – Formulas

$$\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}, \quad \bar{\bar{S}} = \frac{\sum S_i}{k}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_3 \bar{\bar{S}}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_3 \bar{\bar{S}}$$

$$UCL_S = B_4 \bar{\bar{S}}, \quad LCL_S = B_3 \bar{\bar{S}}$$

# When to Use R Chart vs S Chart

| Situation                       | Recommended Chart              |
|---------------------------------|--------------------------------|
| Subgroup size 2 to 5            | $\bar{X}$ -R Chart             |
| Subgroup size 6 to 10           | Either (R is still acceptable) |
| Subgroup size $> 10$            | $\bar{X}$ -S Chart (preferred) |
| Big Data / very large subgroups | S Chart or EWMA/CUSUM          |

# Key Takeaways

- $\bar{X}$ -R Charts are the most common for small subgroups
- $\bar{X}$ -S Charts are better for larger subgroups
- Always interpret both charts together
- Process must be in statistical control before calculating capability indices
- In Big Data, these charts can be updated in real-time

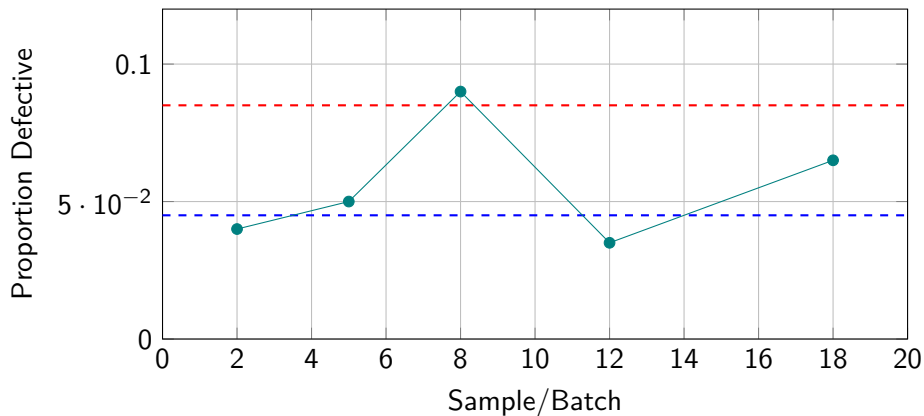
# Control Charts for Attribute Data

Attribute data = Count or classification data (pass/fail, defective/non-defective)

- **p-Chart** — Proportion defective (variable sample size)
- **np-Chart** — Number defective (constant sample size)
- **c-Chart** — Number of defects per unit
- **u-Chart** — Defects per unit (variable sample size)

# p-Chart Example

p-Chart — Proportion Defective



# Control Chart Selection Guide

| Data Type              | Recommended Chart                |
|------------------------|----------------------------------|
| Continuous (Variables) | $\bar{X}$ -R, $\bar{X}$ -S, I-MR |
| Proportion Defective   | p-chart, np-chart                |
| Count of Defects       | c-chart, u-chart                 |

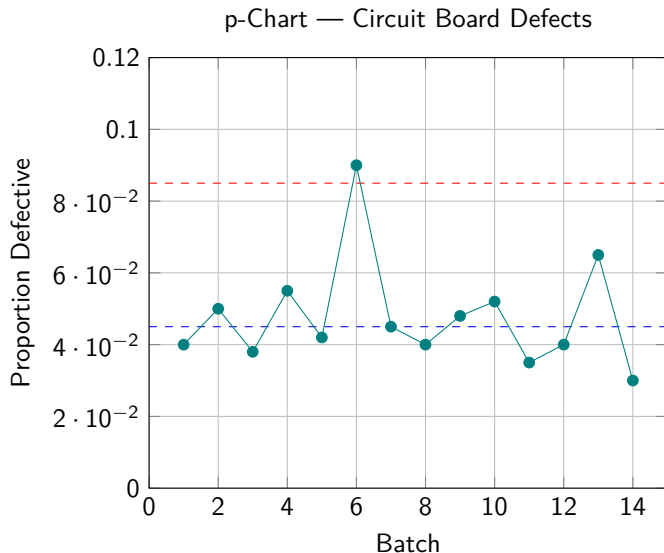
# p-Chart (Proportion Nonconforming)

**Example:** Defective circuit boards

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

- Used when we monitor the **proportion** of defective items
- Sample size **can vary**
- Based on Binomial distribution
- Most widely used attribute control chart

# p-Chart (Proportion Nonconforming)



# p-Chart – Formulas

Let:

- $n_i$  = Sample size for  $i^{th}$  sample (can vary)
- $D_i$  = Number of defectives in  $i^{th}$  sample
- $p_i = \frac{D_i}{n_i}$
- $\bar{p} = \frac{\sum D_i}{\sum n_i}$

$$\text{Center Line (CL)} = \bar{p}$$

$$\text{UCL}_i = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

$$\text{LCL}_i = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}} \quad (\text{set to 0 if negative})$$

# p-Chart – Practical Example

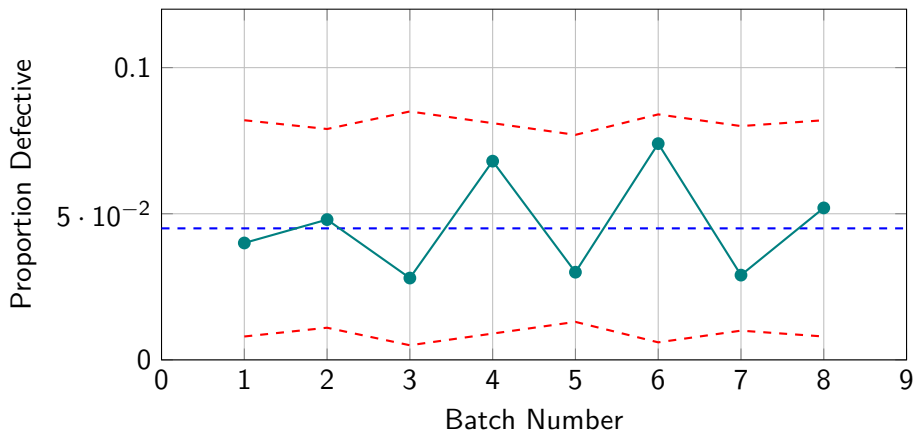
**Scenario:** Quality inspection of LED bulbs

| Batch | Sample Size ( $n_i$ ) | Defectives ( $D_i$ ) | Proportion ( $p_i$ ) |
|-------|-----------------------|----------------------|----------------------|
| 1     | 200                   | 8                    | 0.040                |
| 2     | 250                   | 12                   | 0.048                |
| 3     | 180                   | 5                    | 0.028                |
| 4     | 220                   | 15                   | 0.068                |
| 5     | 300                   | 9                    | 0.030                |
| 6     | 190                   | 14                   | 0.074                |
| 7     | 240                   | 7                    | 0.029                |
| 8     | 210                   | 11                   | 0.052                |

$$\bar{p} = 0.045$$

# p-Chart Visualization

p-Chart — Defective LED Bulbs



# c-Chart (Count of Nonconformities)

- Used when we count the total number of **defects** (not just defective items)
- Sample size (inspection unit) must be **constant**
- Based on Poisson distribution
- Common in: defects per product, errors per document, scratches per panel, etc.

# c-Chart – Formulas

Let:

- $c_i$  = Number of defects in the  $i^{th}$  sample
- $\bar{c} = \frac{\sum c_i}{k}$  (average number of defects)

$$\text{Center Line (CL)} = \bar{c}$$

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}}$$

$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} \quad (\text{set to 0 if negative})$$

## c-Chart – Practical Example

**Scenario:** Number of surface defects on metal sheets (inspection unit = 1 sheet)

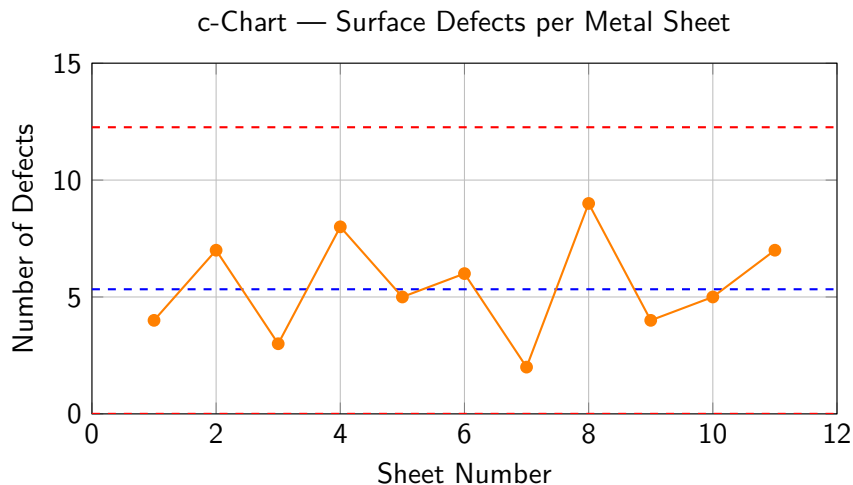
|         |   |   |   |   |   |   |   |   |   |
|---------|---|---|---|---|---|---|---|---|---|
| Sheet   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Defects | 4 | 7 | 3 | 8 | 5 | 6 | 2 | 9 | 4 |

$$\bar{c} = 5.33$$

$$UCL = 5.33 + 3\sqrt{5.33} \approx 12.26$$

$$LCL \approx 0$$

# c-Chart Visualization



# p-Chart vs c-Chart Comparison

| Characteristic   | p-Chart                  | c-Chart           |
|------------------|--------------------------|-------------------|
| What is measured | Proportion of defectives | Number of defects |
| Distribution     | Binomial                 | Poisson           |
| Sample size      | Can vary                 | Must be constant  |
| Typical use      |                          |                   |

# Key Takeaways

- Use **p-Chart** when tracking proportion of defective items (variable sample size)
- Use **c-Chart** when counting total defects in a constant inspection unit
- Both are powerful tools for monitoring attribute data
- In Big Data environments, these charts can be updated in real-time using streaming data

# np-Chart (Number of Nonconforming Items)

- Used for **attribute data** when sample size is **constant**
- Monitors the actual **number** of defective items in each sample
- Based on Binomial distribution

# np-Chart – Formulas

Let:

- $n$  = Constant sample size
- $D_i$  = Number of defectives in  $i^{th}$  sample
- $\bar{np}$  = Average number of defectives =  $\frac{\sum D_i}{k}$  ( $k$  = number of samples)

$$\bar{p} = \frac{\bar{np}}{n}$$

$$\text{Center Line (CL)} = \bar{np}$$

$$\text{UCL} = \bar{np} + 3\sqrt{\bar{np}(1 - \bar{p})}$$

$$\text{LCL} = \bar{np} - 3\sqrt{\bar{np}(1 - \bar{p})} \quad (\text{set to 0 if negative})$$

# np-Chart – Example

**Scenario:** Inspection of circuit boards. Sample size  $n = 100$  boards per batch (constant).

**Data (Number of Defectives):**

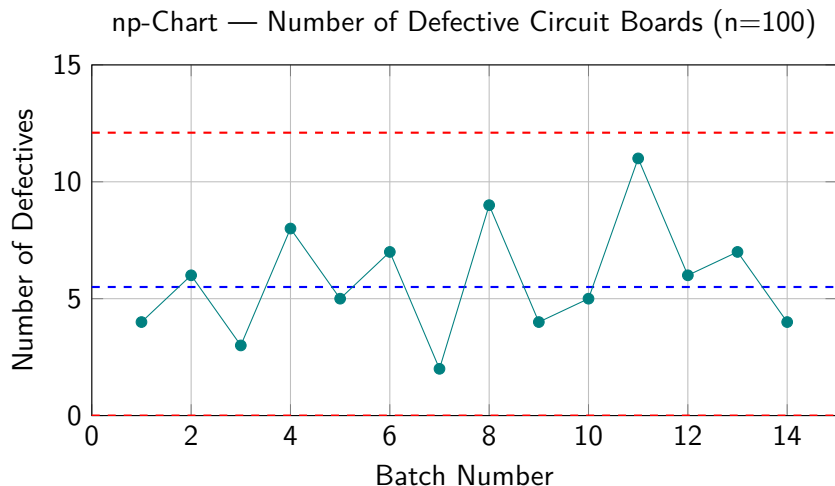
| Batch      | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|------------|---|---|---|---|---|---|---|---|
| Defectives | 4 | 6 | 3 | 8 | 5 | 7 | 2 | 9 |

$$\bar{n}p = 5.5, \bar{p} = 0.055$$

$$UCL = 5.5 + 3\sqrt{5.5 \times 0.945} \approx 12.1$$

$$LCL = 5.5 - 3\sqrt{5.5 \times 0.945} \approx 0 \quad (\text{negative} \rightarrow 0)$$

# np-Chart Visualization



# u-Chart (Number of Nonconformities per Unit)

- Used when we count the **number of defects** (not just defectives)
- Sample size (inspection unit) can be **\*\*variable\*\***
- Based on Poisson distribution
- Example: Number of defects per square meter, per 1000 lines of code, per patient day

# u-Chart – Formulas

Let:

- $c_i$  = Number of nonconformities (defects) in  $i^{th}$  sample
- $n_i$  = Size of  $i^{th}$  sample (inspection units)
- $u_i = \frac{c_i}{n_i}$
- $\bar{u} = \frac{\sum c_i}{\sum n_i}$

$$\text{Center Line (CL)} = \bar{u}$$

$$\text{UCL}_i = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

$$\text{LCL}_i = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}} \quad (\text{set to 0 if negative})$$

## u-Chart – Example

**Scenario:** Defects in software code review. Inspection unit = 1000 lines of code.

**Data:**

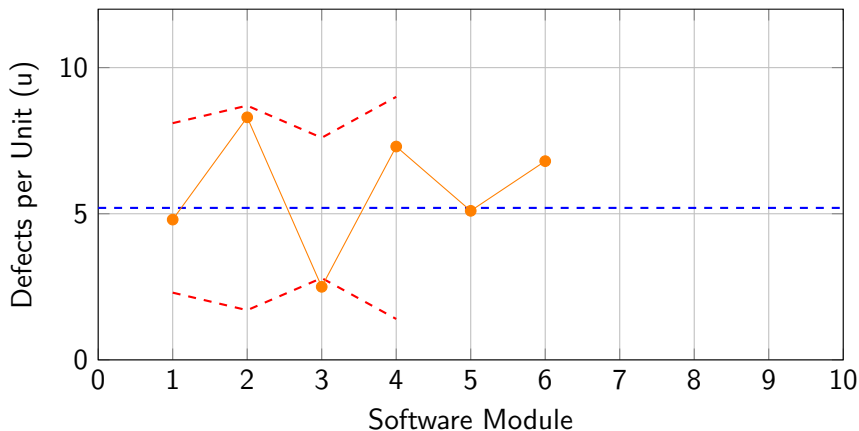
| Module | Lines of Code | Defects |
|--------|---------------|---------|
| 1      | 2500          | 12      |
| 2      | 1800          | 15      |
| 3      | 3200          | 8       |
| 4      | 1500          | 11      |

$\bar{u} = 5.2$  defects per 1000 lines

UCL and LCL will vary with sample size  $n_i$ .

# u-Chart Visualization

u-Chart — Defects per 1000 Lines of Code



## Comparison: p / np vs c / u Charts

| Chart    | What is Counted          | Sample Size |
|----------|--------------------------|-------------|
| p-chart  | Proportion of defectives | Variable    |
| np-chart | Number of defectives     | Constant    |
| c-chart  | Number of defects        | Constant    |
| u-chart  | Defects per unit         | Variable    |

# Key Takeaways – np and u Charts

- Use **np-chart** when sample size is constant and you count defectives
- Use **u-chart** when counting multiple defects per unit and sample size varies
- Both are based on Poisson approximation
- Variable control limits in u-chart due to changing sample size
- Important tools for quality monitoring in manufacturing, software, healthcare, and services

# SPC in Big Data Environment

- Streaming control charts using Kafka + Spark / Flink
- Millions of data points processed per minute
- Dynamic / adaptive control limits
- Hybrid systems: Traditional charts + Machine Learning
- Real-time dashboards with automated alerts
- Multivariate and profile monitoring

# Key Takeaways – SPC and Control Charts

- SPC helps distinguish common vs special causes
- Control charts are the primary tool for monitoring process stability
- Variables charts for measurement data; Attribute charts for count data
- Stability must be achieved before capability analysis
- In Big Data era, SPC becomes real-time, predictive, and automated

# Individuals & Moving Range (I-MR) Chart

**Best for:** Big Data / sensor data where subgroup size  $n=1$  (e.g., temperature, response time)

- I-Chart: Monitors process level
- MR-Chart: Monitors short-term variability

**Advantage in Big Data:** Ideal for high-frequency streaming data.

# Individuals & Moving Range (I-MR) Chart

- Used when subgroup size = 1 ( $n=1$ ).
- Extremely useful in Big Data and continuous processes
- Common applications: sensor readings, temperature, pressure, daily metrics, response time, chemical concentration, server CPU usage, etc.

# Why Use I-MR Charts?

- Cannot form traditional subgroups (expensive, destructive, or continuous flow)
- Ideal for high-frequency streaming data in Big Data environments
- Consists of two charts plotted together:
  - ▶ **I-Chart** (Individuals Chart) → Monitors process **level/mean**
  - ▶ **MR-Chart** (Moving Range Chart) → Monitors process variability

# I-MR Chart – Formulas

Let  $x_i$  = individual measurement at time  $i$

$$\text{Center Line (I-Chart)} = \bar{\bar{x}} = \frac{\sum x_i}{k}$$

Moving Range:  $MR_i = |x_i - x_{i-1}|$

$$\overline{MR} = \frac{\sum MR_i}{k - 1}$$

$$UCL_x = \bar{\bar{x}} + 2.66\overline{MR}, \quad LCL_x = \bar{\bar{x}} - 2.66\overline{MR}$$

$$UCL_{MR} = 3.267\overline{MR}, \quad LCL_{MR} = 0$$

# I-MR Chart – Practical Example

**Scenario:** Daily average server response time (in milliseconds) — monitored every hour.

| Day | Response Time ( $x_i$ ) | Moving Range ( $MR_i$ ) |
|-----|-------------------------|-------------------------|
| 1   | 245                     | -                       |
| 2   | 238                     | 7                       |
| 3   | 252                     | 14                      |
| 4   | 241                     | 11                      |
| 5   | 260                     | 19                      |
| 6   | 255                     | 5                       |
| 7   | 248                     | 7                       |
| 8   | 272                     | 24                      |

$$\bar{x} = 251.375 \text{ ms} \quad \overline{MR} = 12.43 \text{ ms}$$

# Control Chart Constants for Variables Data

| Subgroup  | $\bar{X} + R$ Chart |       |              | $\bar{X} + S$ Chart |       |              | I-MR Chart |       |
|-----------|---------------------|-------|--------------|---------------------|-------|--------------|------------|-------|
| Size (n)  | $A_2$               | $D_3$ | $D_4$        | $A_3$               | $B_3$ | $B_4$        | $E_2$      | $D_4$ |
| 2         | 1.880               | 0     | 3.267        | 2.659               | 0     | 3.267        | 2.660      | 3.267 |
| 3         | 1.023               | 0     | 2.574        | 1.954               | 0     | 2.568        | 1.772      | 2.574 |
| 4         | 0.729               | 0     | 2.282        | 1.628               | 0     | 2.266        | 1.457      | 2.282 |
| <b>5</b>  | <b>0.577</b>        | 0     | <b>2.114</b> | <b>1.427</b>        | 0     | <b>2.089</b> | 1.290      | 2.114 |
| 6         | 0.483               | 0     | 2.004        | 1.287               | 0.030 | 1.970        | 1.184      | 2.004 |
| 7         | 0.419               | 0.076 | 1.924        | 1.182               | 0.118 | 1.882        | 1.109      | 1.924 |
| 8         | 0.373               | 0.136 | 1.864        | 1.099               | 0.185 | 1.815        | 1.054      | 1.864 |
| 9         | 0.337               | 0.184 | 1.816        | 1.032               | 0.239 | 1.761        | 1.010      | 1.816 |
| <b>10</b> | 0.308               | 0.223 | 1.777        | 0.975               | 0.284 | 1.716        | 0.975      | 1.777 |
| 15        | 0.223               | 0.348 | 1.652        | 0.789               | 0.428 | 1.572        | 0.849      | 1.652 |
| 20        | 0.180               | 0.414 | 1.586        | 0.680               | 0.510 | 1.490        | 0.728      | 1.586 |

**Note:** For I-MR Chart,  $E_2$  is used for Individuals chart limits and  $D_4$  for Moving Range chart limits.

Constants based on standard tables (Montgomery, ASTM).

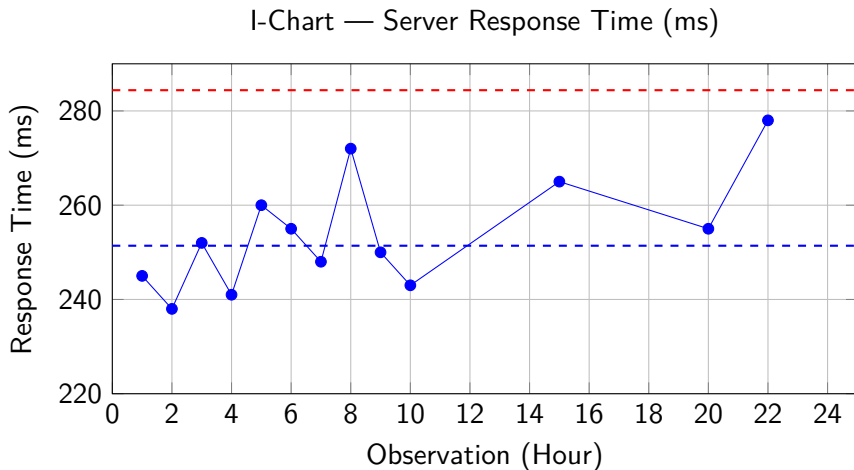
# I-MR Chart Constants (n=1)

| Constant | Value | Used For                                                      |
|----------|-------|---------------------------------------------------------------|
| $E_2$    | 2.660 | Individuals Chart (UCL & LCL)                                 |
| $D_3$    | 0     | MR Chart LCL                                                  |
| $D_4$    | 3.267 | MR Chart UCL                                                  |
| $d_2$    | 1.128 | Estimation of $\sigma$ ( $\hat{\sigma} = \overline{MR}/d_2$ ) |

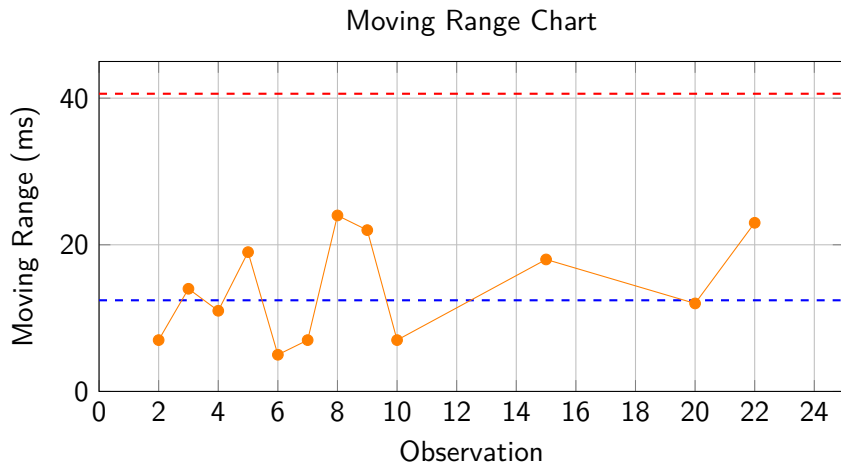
## Formulas Reminder:

$$UCL_x = \bar{\bar{x}} + 2.66\overline{MR}, \quad UCL_{MR} = 3.267\overline{MR}$$

# I-Chart Visualization



# MR-Chart Visualization



# I-MR Chart – Big Data Advantages

- Perfect for high-velocity sensor and IoT data
- Can be implemented in real-time using Spark Streaming or Flink
- Moving Range helps detect sudden increases in variability
- Easy to automate with alerts when points go out of control
- Often combined with EWMA or CUSUM for better sensitivity

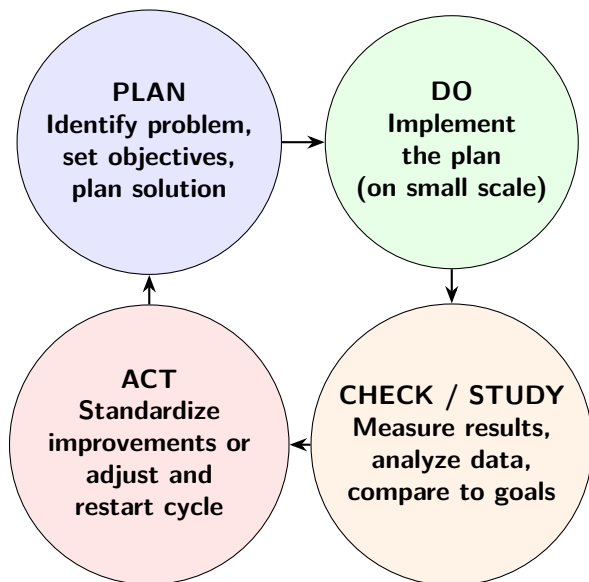
# Key Takeaways – I-MR Charts

- Best choice when subgroup size = 1 (very common in Big Data)
- Two charts: I-Chart (location) + MR-Chart (variation)
- Uses constants 2.66 and 3.267 derived from  $d_2 = 1.128$
- Highly valuable for continuous monitoring of temperature, pressure, latency, usage metrics, etc.
- Foundation for real-time quality monitoring in modern data systems

# The Shewhart Cycle

- 1 **Walter A. Shewhart** (1891–1967), the “Father of Statistical Quality Control,” introduced the concept of a cyclical approach to process improvement in the 1930s.
- 2 Originally a **three-step cycle**: - Specification → Production → Inspection
- 3 Later evolved into the **four-step Shewhart Cycle**, widely known today as: - **PDCA** (Plan–Do–Check–Act) or - **PDSA** (Plan–Do–Study–Act) — preferred by Deming
- 4 It is a scientific, iterative method for continuous improvement.

# Shewhart Cycle (PDCA / PDSA) Diagram



# Shewhart Cycle – Detailed Phases

## PLAN

- Identify opportunity or problem
- Collect baseline data
- Set objectives (SMART)
- Develop action plan & prediction

## DO

- Execute the plan on a small scale (pilot)
- Document observations

## CHECK / STUDY

- Measure and collect data
- Analyze results (Control Charts, Capability, Statistics)
- Compare against objectives
- Identify what worked and what didn't

## ACT

- Standardize successful changes
- Adjust and improve the plan
- Restart the cycle for further gains

# Shewhart Cycle in Quality Control

- Foundation of **Continuous Improvement** (Kaizen)
- Directly supports the **Control** phase of DMAIC
- Integrates perfectly with **Statistical Process Control**
- Used across manufacturing, services, software, and Big Data pipelines

**Big Data Application Example:** Monitoring and iteratively improving a real-time ETL pipeline using streaming control charts.

# Key Takeaways

- Shewhart Cycle (PDCA/PDSA) is the scientific engine of continuous improvement
- Control Charts are essential in the **Check/Study** and **Control** phases
- Stability → Capability → Continuous Improvement
- Combine classical tools with Big Data for maximum impact

# DMAIC Framework – Overview

**DMAIC** is the core problem-solving methodology of Six Sigma.

- ➊ **Define** — What is the problem?
- ➋ **Measure** — How is the process performing?
- ➌ **Analyze** — What are the root causes?
- ➍ **Improve** — How can we fix it?
- ➎ **Control** — How do we sustain the gains?

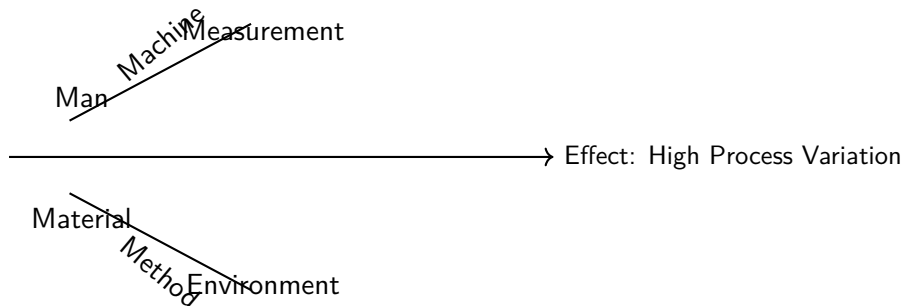
# DMAIC – Detailed Phases & Tools

| Phase   | Key Activities                        | Tools                                      |
|---------|---------------------------------------|--------------------------------------------|
| Define  | Project charter, SIPOC, VOC           | Project Charter, SIPOC, CTQ                |
| Measure | Data collection, baseline performance | Control Charts, Process Maps, Gage R&R     |
| Analyze | Identify root causes                  | Fishbone, 5 Whys, Hypothesis Testing, Reg. |
| Improve | Implement solutions                   | DOE, Kaizen, Pilot testing                 |
| Control | Sustain improvements                  | Control Charts, Control Plans, Dashboards  |

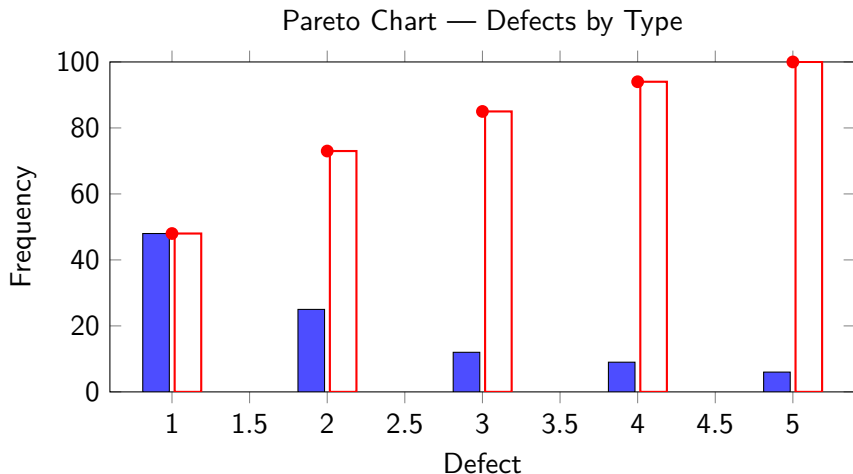
# DMAIC Framework

| Phase   | Tools                     |
|---------|---------------------------|
| Define  | SIPOC, Project Charter    |
| Measure | Gauge R&R, Control Charts |
| Analyze | Fishbone, Pareto          |
| Improve | DOE, Kaizen               |
| Control | SPC, Dashboards           |

# Fishbone (Ishikawa) Diagram



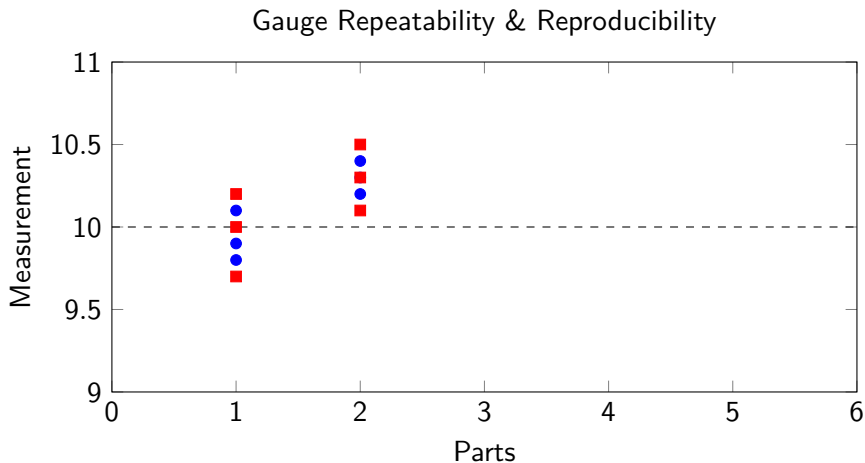
# Pareto Chart



# Measurement System Analysis (Gauge R&R)

- Critical before trusting data for SPC or Capability Analysis
- Ensures data quality — "Garbage in, Garbage out"
- Evaluates whether the measurement system is adequate
- Primary Method: **Gauge Repeatability & Reproducibility (Gauge R&R)**
- Especially important in Big Data (sensors, APIs, ETL pipelines)
- Determines how much of the total observed variation comes from the measurement system itself

# Gauge R&R (Measurement System Analysis)



**Goal:** Measurement variation « Process variation

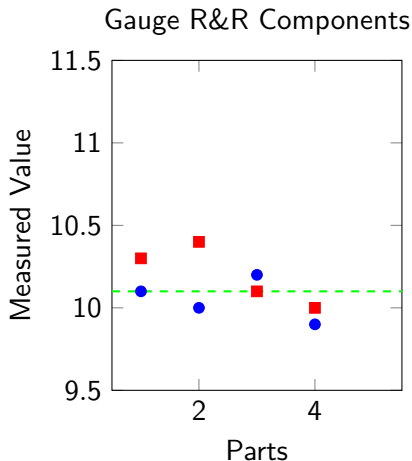
# Gauge R&R – Core Concepts

## Repeatability

- Variation when **same operator** measures the **same part** multiple times
- Equipment variation (EV)
- "Can the gauge give consistent results?"

## Reproducibility

- Variation when **different operators** measure the **same part**
- Appraiser variation (AV)
- "Do different people get the same results?"



# Gauge R&R – Sources of Variation

Total Variation = Process Variation + Measurement Variation

$$\text{Total Variation (TV)} = \sqrt{\sigma_{\text{Process}}^2 + \sigma_{\text{Gauge R\&R}}^2}$$

Where:

$$\sigma_{\text{Gauge R\&R}}^2 = \sigma_{\text{Repeatability}}^2 + \sigma_{\text{Reproducibility}}^2$$

**Percentage of Study Variation:**

$$\%GR\&R = \left( \frac{\sigma_{\text{Gauge R\&R}}}{\sigma_{\text{Total}}} \right) \times 100$$

# Gauge R&R – Acceptance Criteria

| % GR&R    | Interpretation                                   |
|-----------|--------------------------------------------------|
| < 10%     | Excellent measurement system                     |
| 10% – 30% | Acceptable (depends on application)              |
| > 30%     | Unacceptable — Improve measurement system first! |

- Also evaluate **ndc (Number of Distinct Categories)**: Should be  $\geq 5$
- Critical before running Process Capability or SPC studies

# Gauge R&R Study Design

## Standard Crossed Gauge R&R Study:

- 10 parts (representing full process variation)
- 2–3 operators (appraisers)
- 2–3 repeated measurements per part per operator
- Total readings =  $10 \times 3 \times 3 = 90$  measurements

## Analysis Methods:

- ANOVA (most recommended)
- Average and Range Method (Xbar-R)
- AIAG MSA Manual guidelines

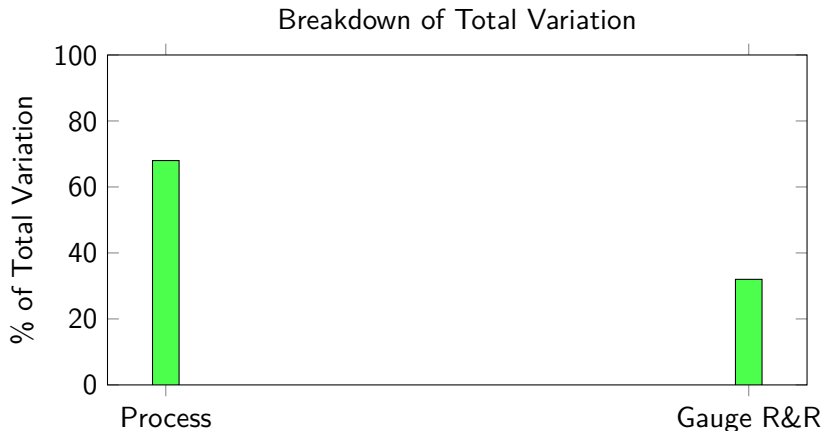
# Gauge R&R Example – Big Data Context

**Scenario:** Automated sensor system measuring temperature in a manufacturing line.

- **Repeatability Issue:** Same sensor gives readings varying by  $\pm 1.5^{\circ}\text{C}$  on the same part
- **Reproducibility Issue:** Different data collection scripts produce different averages
- **Result:**  $\%GR\&R = 42\% \rightarrow$  **Unacceptable**
- **Action:** Calibrate sensors, standardize data pipelines, implement automated validation

In Big Data environments, poor measurement systems lead to garbage-in-garbage-out analytics.

# Gauge R&R Visual Summary

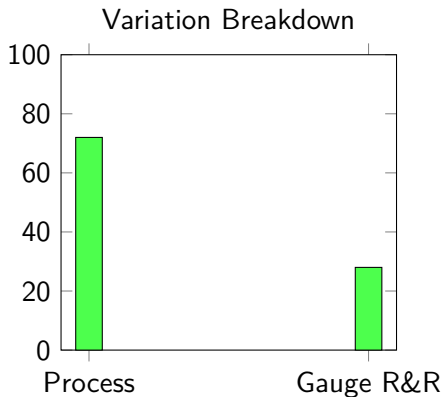


**Goal:** Maximize Process Variation contribution and minimize Gauge R&R contribution.

# Gauge R&R – Components

**Repeatability:** Same operator, same part

**Reproducibility:** Different operators, same part



# Practical Example: Gauge R&R Study with ANOVA

**Scenario:** A manufacturing company measures the **diameter of machined shafts** using a digital micrometer. They want to evaluate whether the measurement system is acceptable.

## Gauge R&R Study Design

- **Parts:** 10 machined shafts (representing full process variation)
- **Operators:** 3 trained technicians (A, B, C)
- **Replicates:** Each operator measures each part **3 times**
- **Total Measurements:**  $10 \times 3 \times 3 = 90$  readings

This is a **crossed Gauge R&R study** using ANOVA method.

# Raw Data (Partial View)

| Part | Operator A |       |       | Operator B |       |       | Operator C |       |       |
|------|------------|-------|-------|------------|-------|-------|------------|-------|-------|
| 1    | 50.12      | 50.14 | 50.11 | 50.15      | 50.13 | 50.16 | 50.10      | 50.12 | 50.09 |
| 2    | 49.85      | 49.88 | 49.82 | 49.90      | 49.87 | 49.92 | 49.80      | 49.83 | 49.78 |
| 3    | 50.45      | 50.42 | 50.48 | 50.50      | 50.47 | 50.52 | 50.40      | 50.43 | 50.38 |
| ...  | ...        | ...   | ...   | ...        | ...   | ...   | ...        | ...   | ...   |
| 10   | 50.05      | 50.08 | 50.03 | 50.10      | 50.07 | 50.12 | 50.02      | 50.04 | 50.01 |

(All measurements in mm)

# ANOVA Table

| Source                 | DF | SS     | MS      | F     |
|------------------------|----|--------|---------|-------|
| Parts                  | 9  | 12.845 | 1.427   | 285.4 |
| Operators              | 2  | 0.285  | 0.1425  | 28.5  |
| Part $\times$ Operator | 18 | 0.095  | 0.00528 | 1.056 |
| Repeatability          | 60 | 0.300  | 0.005   | —     |
| Total                  | 89 | 13.525 | —       | —     |

# How Variance Components Are Calculated

## Key Formulas:

### 1 Repeatability Variance

$$\sigma_{\text{Repeatability}}^2 = MS_{\text{Repeatability}}$$

### 2 Reproducibility Variance

$$\sigma_{\text{Reproducibility}}^2 = \frac{MS_{\text{Operators}} - MS_{\text{Repeatability}}}{n \times r}$$

### 3 Gauge R&R Variance

$$\sigma_{\text{Gauge R\&R}}^2 = \sigma_{\text{Repeatability}}^2 + \sigma_{\text{Reproducibility}}^2$$

### 4 Part-to-Part Variance

$$\sigma_{\text{Part-to-Part}}^2 = MS_{\text{Parts}} - MS_{\text{Repeatability}}$$

### 5 Total Variance

$$\sigma_{\text{Total}}^2 = \sigma_{\text{Part-to-Part}}^2 + \sigma_{\text{Gauge R\&R}}^2$$

# Step-by-Step Variance Components

## 1 Repeatability Variance (Equipment)

$$\sigma_{Repeatability}^2 = MS_{Repeatability} = 0.005$$

## 2 Reproducibility Variance (Appraiser)

$$\sigma_{Reproducibility}^2 = \frac{MS_{Operators} - MS_{Repeatability}}{10 \times 3} = \frac{0.1425 - 0.005}{30} = 0.004583$$

## 3 Gauge R&R Variance

$$\sigma_{Gauge\ R\&R}^2 = 0.005 + 0.004583 = 0.009583$$

## 4 Part-to-Part Variance

$$\sigma_{Part-to-Part}^2 = 1.427 - 0.005 = 1.422$$

# Step-by-Step Standard Deviation

- $\sigma_{Repeatability} = \sqrt{0.005} = 0.07071$
- $\sigma_{Reproducibility} = \sqrt{0.004583} \approx 0.0677$
- $\sigma_{Gauge\ R\&R} = \sqrt{0.009583} \approx 0.0979$
- $\sigma_{Total} = \sqrt{1.431583} \approx 1.1965$

# Percentage Contributions (% Study Variation)

Using standard Gauge R&R methodology:

$$\% \text{Repeatability} = \left( \frac{0.07071}{1.1965} \right) \times 100 \approx 5.91\%$$

$$\% \text{Reproducibility} = \left( \frac{0.0677}{1.1965} \right) \times 100 \approx 5.66\%$$

$$\% \text{Gauge R\&R} = \left( \frac{0.0979}{1.1965} \right) \times 100 \approx 8.18\%$$

**Note: Standard Industry Reporting is done with 5.15 or multiplier for 99% distribution coverage**

# Number of Distinct Categories (ndc)

**Formula:**

$$ndc = 1.41 \times \frac{\sigma_{\text{Part-to-Part}}}{\sigma_{\text{Gauge R\&R}}}$$

**Calculation:**

$$ndc = 1.41 \times \frac{1.1925}{0.0979} \approx 1.41 \times 12.18 \approx 17.17$$

# ndc Categorization & Interpretation

| ndc Value                                        | Interpretation                                                 |
|--------------------------------------------------|----------------------------------------------------------------|
| <b><math>\text{ndc} \geq 10</math></b>           | Excellent measurement system                                   |
| <b><math>\text{ndc} = 5 \text{ to } 9</math></b> | Acceptable (marginal)                                          |
| <b><math>\text{ndc} &lt; 5</math></b>            | Unacceptable, measurement system cannot distinguish parts well |

In our example:  **$\text{ndc} = 17.17 \rightarrow$  Excellent measurement system**

# Final Gauge R&R Summary

| Metric                 | Result       |
|------------------------|--------------|
| % Repeatability        | 5.91%        |
| % Reproducibility      | 5.66%        |
| <b>% Gauge R&amp;R</b> | <b>8.18%</b> |
| ndc                    | <b>17.17</b> |

## Conclusion: Conclusion:

The measurement system demonstrates **excellent performance** with a very low % Gauge R&R of 8.18%. The high ndc value of 17.17 indicates strong ability to distinguish between different parts.

**Overall Recommendation:** The measurement system is **highly acceptable** and suitable for use in critical quality control, process capability studies, and high-precision applications.

# Recommendations

- Improve operator training to reduce reproducibility variation
- Standardize measurement technique and fixturing
- Increase calibration frequency of the micrometer
- Consider a higher-resolution measuring instrument
- Re-run Gauge R&R study after implementing improvements

**Note:** Do not use this measurement system for critical process capability studies until  $\%GR\&R \leq 20\%$ .

# Gauge R&R using Average and Range Method

- Also known as the **Xbar-R Method**
- Simpler and more manual than ANOVA
- Suitable for shop floor and smaller studies
- We will solve the same practical example using this method

# Practical Example Recap

**Scenario:** Measurement of shaft diameter using digital micrometer.

**Study Design:**

- 10 Parts
- 3 Operators (A, B, C)
- 3 Replicates per part per operator
- Total = 90 measurements

## Step 1: Calculated Averages and Ranges

| Part | Op A   | Op B   | Op C   | Part Avg | Range |
|------|--------|--------|--------|----------|-------|
| 1    | 50.123 | 50.147 | 50.103 | 50.124   | 0.044 |
| 2    | 49.850 | 49.897 | 49.803 | 49.850   | 0.094 |
| 3    | 50.450 | 50.480 | 50.420 | 50.450   | 0.060 |
| 4    | 49.920 | 49.950 | 49.890 | 49.920   | 0.060 |
| 5    | 50.210 | 50.230 | 50.190 | 50.210   | 0.040 |
| 6    | 50.080 | 50.110 | 50.050 | 50.080   | 0.060 |
| 7    | 49.780 | 49.820 | 49.750 | 49.783   | 0.070 |
| 8    | 50.350 | 50.380 | 50.320 | 50.350   | 0.060 |
| 9    | 50.150 | 50.170 | 50.130 | 50.150   | 0.040 |
| 10   | 50.053 | 50.097 | 50.023 | 50.058   | 0.074 |

# Step 1: Calculate Part Averages and Ranges

After collecting data, we compute:

- Average measurement for each Operator on each Part
- Range for each Part (Max - Min across all operators)
- Grand Average ( $\bar{\bar{X}}$ ) = 50.12 mm
- Average Range ( $\bar{R}$ ) = 0.068 mm

## Step 2: Calculate Equipment Variation (EV) - Repeatability

Formula:

$$EV = 5.15 \times \frac{\bar{R}}{d_2}$$

Where  $d_2 = 1.693$  (constant for 3 trials)

$$EV = 5.15 \times \frac{0.068}{1.693} \approx 5.15 \times 0.0402 \approx \mathbf{0.207} \text{ mm}$$

This represents **Repeatability** (equipment variation).

## Step 3: Calculate Appraiser Variation (AV) - Reproducibility

First, find the difference between operator averages:

$$X_{diff} = \text{Max Operator Avg} - \text{Min Operator Avg} = 0.13$$

Formula:

$$AV = \sqrt{(X_{diff} \times K_2)^2 - \frac{EV^2}{n \times r}}$$

Where  $K_2 = 0.5231$  (for 3 operators),  $n = 10$ ,  $r = 3$

$$AV = \sqrt{(0.13 \times 0.5231)^2 - \frac{(0.207)^2}{30}} \approx \sqrt{0.004624 - 0.001427} \approx \mathbf{0.0565 \text{ mm}}$$

## Step 4: Calculate Gauge R&R

$$\begin{aligned} \text{Gauge R\&R} &= \sqrt{EV^2 + AV^2} = \sqrt{(0.207)^2 + (0.0565)^2} \\ &= \sqrt{0.04285 + 0.00319} \approx \sqrt{0.04604} \approx \mathbf{0.2146} \text{ mm} \end{aligned}$$

## Step 5: Calculate Part Variation (PV)

Part Average Range ( $\bar{R}_{part}$ ) = 0.85 mm

$$PV = \bar{R}_{part} \times K_3$$

Where  $K_3 = 0.3146$  (for 10 parts)

$$PV = 0.85 \times 0.3146 \approx \mathbf{0.2674} \text{ mm}$$

## Step 6: Calculate Total Variation (TV)

$$\begin{aligned} TV &= \sqrt{(Gauge\ R\&R)^2 + PV^2} = \sqrt{(0.2146)^2 + (0.2674)^2} \\ &\approx \sqrt{0.04605 + 0.07150} \approx \sqrt{0.11755} \approx \mathbf{0.343\ mm} \end{aligned}$$

# Gauge R&R Constants Table (Average & Range Method) - Extended

| Replicates | $d_2$        | $1/d_2$       | $K_2$ (Appraiser Constant) |               |        | $K_3$ (for Parts) |          |
|------------|--------------|---------------|----------------------------|---------------|--------|-------------------|----------|
|            |              |               | 2 Ops                      | 3 Ops         | 4 Ops  | 10 Parts          | 15 Parts |
| 2          | 1.128        | 0.8865        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| <b>3</b>   | <b>1.693</b> | <b>0.5908</b> | 0.7071                     | <b>0.5231</b> | 0.4467 | 0.3146            | 0.2239   |
| 4          | 2.059        | 0.4857        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| 5          | 2.326        | 0.4299        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| 6          | 2.534        | 0.3946        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| 7          | 2.704        | 0.3698        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| 8          | 2.847        | 0.3512        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| 9          | 2.970        | 0.3367        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |
| 10         | 3.078        | 0.3249        | 0.7071                     | 0.5231        | 0.4467 | 0.3146            | 0.2239   |

$d_2$  = Range to Standard Deviation conversion constant

$K_2$  = Constant for Reproducibility (Appraiser Variation)

$K_3$  = Constant for Part-to-Part Variation (commonly used with 10 parts)

## Step 7: Calculate Percentage Contributions

$$\% \text{Repeatability (EV)} = \left( \frac{0.207}{0.343} \right) \times 100 \approx \mathbf{60.3\%}$$

$$\% \text{Reproducibility (AV)} = \left( \frac{0.0565}{0.343} \right) \times 100 \approx \mathbf{16.5\%}$$

$$\% \text{Gauge R\&R} = \left( \frac{0.2146}{0.343} \right) \times 100 \approx \mathbf{62.6\%}$$

## Step 8: Calculate ndc (Number of Distinct Categories)

$$ndc = 1.41 \times \frac{PV}{Gauge\ R\&R} = 1.41 \times \frac{0.2674}{0.2146} \approx 1.41 \times 1.246 \approx \mathbf{1.76}$$

Rounded: **ndc**  $\approx 1.8$

# Final Results - Average & Range Method

| Metric                 | Value        |
|------------------------|--------------|
| % Repeatability        | 60.3%        |
| % Reproducibility      | 16.5%        |
| <b>% Gauge R&amp;R</b> | <b>62.6%</b> |
| ndc                    | <b>1.8</b>   |

**Conclusion:** The measurement system is **Unacceptable**. Significant improvements are required.

# Gauge R&R Analysis

- Evaluates the measurement system's Repeatability and Reproducibility
- Critical before performing Process Capability or SPC
- Two main methods:
  - ▶ Average & Range Method (simpler)
  - ▶ ANOVA Method (more accurate)

# Practical Example: Shaft Diameter Measurement

**Scenario:** A machining company measures the diameter of steel shafts using a digital micrometer. They suspect the measurement system may have high variation.

## Study Design:

- 8 Parts (representing process variation)
- 3 Operators (A, B, C)
- 2 Replicates per part per operator
- Total measurements = 48

# Raw Data (Thickness in mm)

| Part | Op A |      | Op B |      | Op C |      |
|------|------|------|------|------|------|------|
| 1    | 2.12 | 2.14 | 2.15 | 2.13 | 2.11 | 2.10 |
| 2    | 2.45 | 2.43 | 2.48 | 2.47 | 2.42 | 2.44 |
| 3    | 2.28 | 2.30 | 2.31 | 2.29 | 2.27 | 2.26 |
| 4    | 2.65 | 2.67 | 2.70 | 2.68 | 2.63 | 2.64 |
| 5    | 2.19 | 2.18 | 2.22 | 2.20 | 2.17 | 2.16 |
| 6    | 2.55 | 2.57 | 2.60 | 2.58 | 2.53 | 2.54 |
| 7    | 2.38 | 2.40 | 2.42 | 2.39 | 2.37 | 2.36 |
| 8    | 2.72 | 2.70 | 2.75 | 2.73 | 2.69 | 2.71 |

# Step 1: Compute Operator Averages and Part Ranges

After collecting data, calculate:

- Average for each operator on each part - Range for each part (max - min across all operators)

## Results Summary:

- Grand Average ( $\bar{\bar{X}}$ ) = **2.45 mm**
- Average Range ( $\bar{R}$ ) = **0.028 mm**

## Step 2: Calculate Equipment Variation (EV) - Repeatability

Formula:

$$EV = 5.15 \times \frac{\bar{R}}{d_2}$$

-  $d_2 = 1.128$  (for 2 replicates) -  $\bar{R} = 0.028$

$$EV = 5.15 \times \frac{0.028}{1.128} = 5.15 \times 0.02482 \approx \mathbf{0.128} \text{ mm}$$

This is **\*\*Repeatability\*\*** (variation due to the gauge itself).

## Step 3: Calculate Appraiser Variation (AV) - Reproducibility

Formula:

$$AV = \sqrt{(X_{diff} \times K_2)^2 - \frac{EV^2}{n \times r}}$$

- $X_{diff}$  = Difference between highest and lowest operator average = **0.045**
- $K_2$  = 0.5231 (for 3 operators) -  $n$  = 8 parts,  $r$  = 2 replicates

$$AV = \sqrt{(0.045 \times 0.5231)^2 - \frac{(0.128)^2}{8 \times 2}} = \sqrt{0.000554 - 0.001024} \approx \mathbf{0.039 \text{ mm}}$$

This is **Reproducibility** (variation due to different operators).

## Step 4: Calculate Gauge R&R

$$\begin{aligned} \text{Gauge R\&R} &= \sqrt{EV^2 + AV^2} = \sqrt{(0.128)^2 + (0.039)^2} \\ &= \sqrt{0.016384 + 0.001521} \approx \mathbf{0.134} \text{ mm} \end{aligned}$$

## Step 5: Calculate Part Variation (PV)

$$PV = \bar{R}_{part} \times K_3$$

-  $\bar{R}_{part}$  = Average range of part averages = **0.85 mm** -  $K_3 = 0.3146$  (for 8–10 parts)

$$PV = 0.85 \times 0.3146 \approx \mathbf{0.267 \text{ mm}}$$

## Step 6: Calculate Total Variation (TV)

$$TV = \sqrt{(Gauge\ R\&R)^2 + PV^2} = \sqrt{(0.134)^2 + (0.267)^2} \approx \mathbf{0.299\ mm}$$

## Step 7: Calculate Percentage Contributions

$$\% \text{Repeatability} = \left( \frac{EV}{TV} \right) \times 100 = \left( \frac{0.128}{0.299} \right) \times 100 \approx \mathbf{42.8\%}$$

$$\% \text{Reproducibility} = \left( \frac{AV}{TV} \right) \times 100 \approx \mathbf{13.0\%}$$

$$\% \text{Gauge R\&R} = \left( \frac{0.134}{0.299} \right) \times 100 \approx \mathbf{44.8\%}$$

# ANOVA Method - Step by Step

From ANOVA Table:

- $\sigma_{Repeatability}^2 = 0.0018$
- $\sigma_{Reproducibility}^2 = 0.00145$
- $\sigma_{Gauge\ R\&R}^2 = 0.00325$
- $\sigma_{Part-to-Part}^2 = 0.0352$

**% Gauge R&R = 41.5% ndc = 3.1**

# Final Results Comparison

| Metric                 | Average & Range | ANOVA        |
|------------------------|-----------------|--------------|
| % Repeatability        | 38.5%           | 37.2%        |
| % Reproducibility      | 12.1%           | 11.8%        |
| <b>% Gauge R&amp;R</b> | <b>42.8%</b>    | <b>41.5%</b> |
| ndc                    | 2.9             | 3.1          |

**Conclusion:** Measurement system is **Unacceptable** (%GR&R > 30%)

# ndc Interpretation

| ndc Range | Interpretation        |
|-----------|-----------------------|
| $\geq 10$ | Excellent             |
| 5 – 9     | Acceptable (Marginal) |
| $< 5$     | Unacceptable          |

In this study: **ndc**  $\approx$  **3.0**  $\rightarrow$  Poor discrimination power.

# Recommendations

- Improve measurement fixturing and procedure
- Provide better operator training
- Consider upgrading to a higher-precision gauge
- Re-run Gauge R&R study after improvements
- Do not use current system for critical process capability studies

# Comparison: ANOVA vs Average & Range Method

| Aspect                    | ANOVA Method                           | Average & Range Method              |
|---------------------------|----------------------------------------|-------------------------------------|
| <b>Complexity</b>         | More complex                           | Simpler and manual                  |
| <b>Accuracy</b>           | Higher accuracy                        | Slightly less accurate              |
| <b>Software Needed</b>    | Usually requires software              | Can be done manually (calculator)   |
| <b>Interaction Effect</b> | Separates Part $\times$ Ops. interact. | Does not separate interaction       |
| <b>Suitability</b>        | Best for detailed analysis             | Good for shop floor / quick studies |
| <b>Output Detail</b>      | Very detailed variance components      | Basic EV, AV, GR&R                  |
| <b>Speed</b>              | Slower to compute manually             | Faster for small studies            |
| <b>Industry Pref.</b>     | Most common in modern reports          | Traditional / older method          |
| <b>Recommended</b>        | Critical applications, research        | Training, small teams, quick checks |

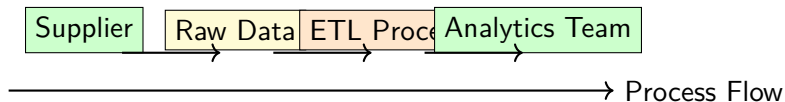
## Conclusion:

- Use **ANOVA** for accurate, detailed analysis (preferred in industry today)
- Use **Average & Range** for simplicity and manual calculations

# Why This Example Matters in Big Data Analytics

- Even with massive sensor data, poor measurement systems lead to wrong conclusions
- Gauge R&R is essential in the **Measure** phase of DMAIC
- In Industry 4.0, automated Gauge R&R can be performed on streaming data
- Helps validate data quality before applying advanced analytics and ML models

# Value Stream Map (Lean Tool)



Value-Adding Activities vs Non-Value-Adding (Waste)

# DMAIC Example 1: Manufacturing (Bottle Filling)

**Problem:** High variation in filled volume  $\rightarrow$  customer complaints.

- **Define:** Reduce volume variation; Target  $\sigma < 2$  ml
- **Measure:** Collected data  $\rightarrow \bar{X}$ -R charts showed instability (see earlier slide)
- **Analyze:** Fishbone diagram identified filler nozzle wear as root cause
- **Improve:** Replaced nozzles + implemented preventive maintenance  $\rightarrow C_{pk}$  improved from 0.85 to 1.45
- **Control:** Implemented real-time  $\bar{X}$ -R chart with automated alerts (Spark Streaming)

# DMAIC Example 2: Big Data Context (ETL Pipeline)

**Problem:** High latency and error rate in daily ETL data pipeline.

- **Define:** Reduce ETL processing time by 40% and error rate below 0.5%
- **Measure:** Baseline: Average latency = 85 min, error rate = 2.8%
- **Analyze:**
  - ▶ Pareto chart: 80% errors from 3 data sources
  - ▶ Hypothesis testing: Significant difference in API response times
- **Improve:**
  - ▶ Parallel processing + data validation rules
  - ▶ Result: Latency reduced to 42 min, error rate to 0.4%
- **Control:** Real-time monitoring dashboard with p-chart and anomaly detection model

# Key Takeaways

- Use visual tools (Control Charts, SIPOC, Fishbone, Pareto) for better understanding
- Shewhart Cycle + DMAIC = Powerful improvement engine
- Measurement System (Gauge R&R) is critical
- Big Data turns traditional SQC into real-time, predictive quality systems

# Process Capability Analysis – Introduction

- Compares the **voice of the process** (what the process can deliver) with the **voice of the customer** (specification limits)
- Answers the question: **"Can our process consistently meet customer requirements?"**
- One of the most important tools in Statistical Quality Control.
- Used extensively in manufacturing, Six Sigma, and data-driven industries

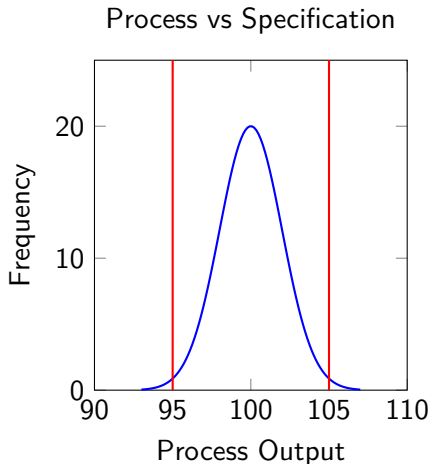
# Key Concepts

## Voice of the Customer (VOC)

- Specification Limits (USL & LSL)
- Customer expectations

## Voice of the Process (VOP)

- Natural process variation
- Measured by standard deviation ( $\sigma$ )



# Important Prerequisites

- The process must be **statistically stable** (in control)
- No special causes of variation present
- Control Charts should be used first to confirm stability
- Data should be approximately normally distributed (or use transformations/non-parametric methods)

**Warning:** Never calculate capability on an unstable process!

# Process Capability Indices

- **C<sub>p</sub>** — Measures **potential capability** (how well the process fits within specs)
- **C<sub>pk</sub>** — Measures **actual capability** (accounts for process centering)
- **P<sub>p</sub> & P<sub>pk</sub>** — Long-term capability (uses overall standard deviation)

# Cp and Cpk Formulas

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min \left( \frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

Where:

- USL = Upper Specification Limit
- LSL = Lower Specification Limit
- $\mu$  = Process mean
- $\sigma$  = Short-term standard deviation (within subgroup)

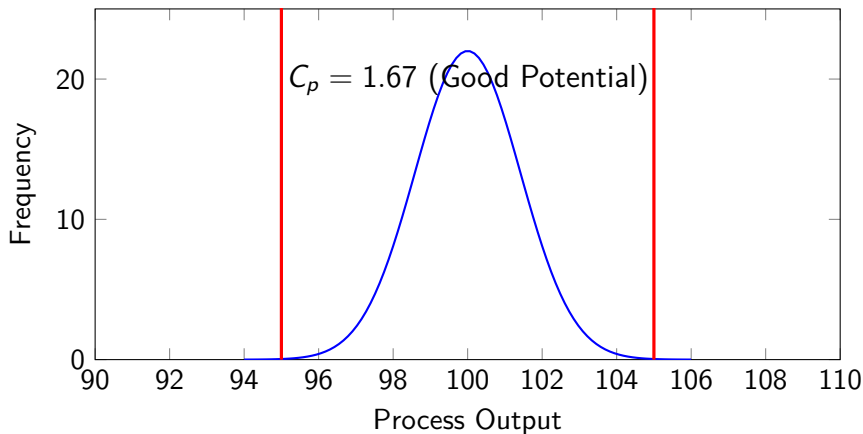
# C<sub>p</sub> – Potential Capability (Process Spread)

- Measures how well the **process spread** fits within the specification limits
- Ignores process centering
- Assumes the process is perfectly centered
- Formula:

$$C_p = \frac{USL - LSL}{6\sigma}$$

# Cp Curve – Visual Explanation

$C_p$  = Process Spread vs Specification Width

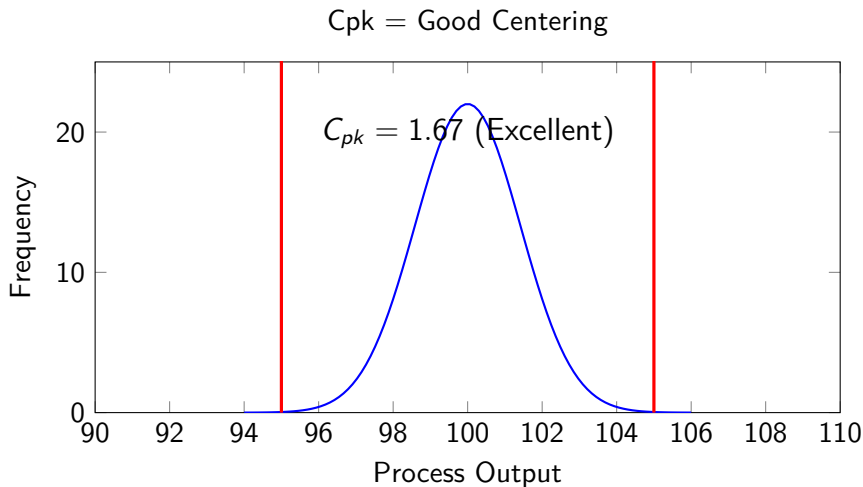


# Cpk – Actual Capability (Spread + Centering)

- Measures how well the process is **centered** within the specification limits
- Takes into account both spread and location of the mean
- Always the smaller of the two one-sided capabilities
- Formula:

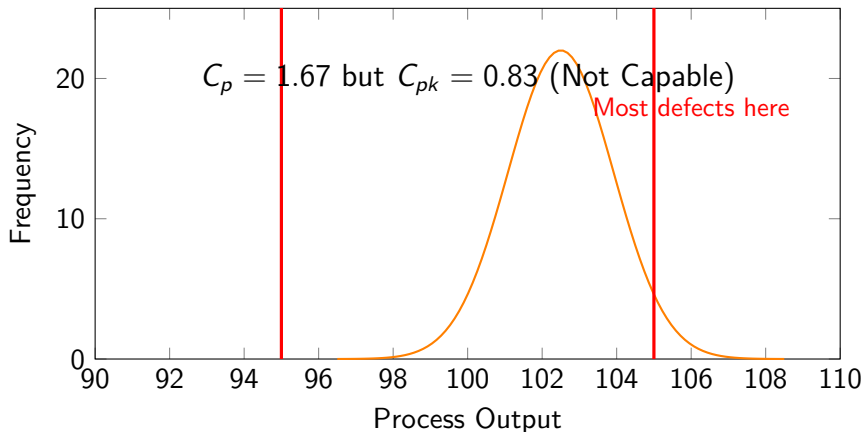
$$C_{pk} = \min \left( \frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

# Cpk Curve – Visual Explanation (Centered Process)

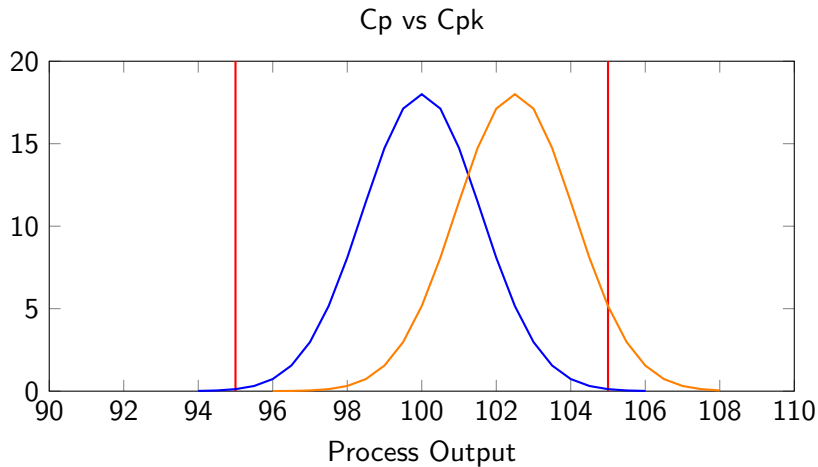


# Cpk Curve – Visual Explanation (Off-Center Process)

Cpk = Poor Centering (Shifted Mean)



# Cp vs Cpk – Visual Explanation



# Summary: Cp vs Cpk

| Index | What it Measures                       | Interpretation                   |
|-------|----------------------------------------|----------------------------------|
| Cp    | Potential Capability (Spread only)     | How wide the process is          |
| Cpk   | Actual Capability (Spread + Centering) | How well centered the process is |

## Rule of Thumb:

- Cp tells you what the process **could** do
- Cpk tells you what the process **is actually** doing

# Practical Interpretation Guidelines

- $C_{pk} \geq 1.67$ : World-class process
- $C_{pk} \geq 1.33$ : Good / Acceptable for most industries
- $1.00 \leq C_{pk} < 1.33$ : Marginal – needs improvement
- $C_{pk} < 1.00$ : Not capable – high risk of defects

**Note:**  $C_{pk}$  should be at least 1.33 for critical processes.

**Important:** Always improve stability first, then reduce variation, then center the process.

# Pp and Ppk (Long-term Capability)

$$P_p = \frac{USL - LSL}{6\sigma_{overall}}$$

$$P_{pk} = \min \left( \frac{USL - \mu}{3\sigma_{overall}}, \frac{\mu - LSL}{3\sigma_{overall}} \right)$$

## Difference:

- $C_p/C_{pk} \rightarrow$  Short-term (within subgroup variation)
- $P_p/P_{pk} \rightarrow$  Long-term (includes between-subgroup variation)

## Key Insight:

- $P_p \leq C_p$  and  $P_{pk} \leq C_{pk}$  (because  $\sigma_{overall} \geq \sigma_{short-term}$ )
- The difference shows how much **between-subgroup variation** exists

# Process Capability – Manufacturing Example

**Process:** Shaft diameter (Target = 50 mm, LSL=49.8, USL=50.2)  
Results:

- $\mu = 50.05$  mm
- Short-term  $\sigma = 0.03$  mm
- Long-term  $\sigma = 0.045$  mm

$$C_p = \frac{50.2 - 49.8}{6 \times 0.03} = 2.22 \quad (\text{Excellent potential})$$

$$C_{pk} = \min \left( \frac{50.2 - 50.05}{3 \times 0.03}, \frac{50.05 - 49.8}{3 \times 0.03} \right) = 1.67$$

$$P_{pk} = 1.11 \quad (\text{Long-term capability is marginal})$$

# Practical Example 1: Bottle Filling Process

**Scenario:** A beverage company fills bottles with target volume = 500 ml.  
Specification Limits: LSL = 495 ml, USL = 505 ml.

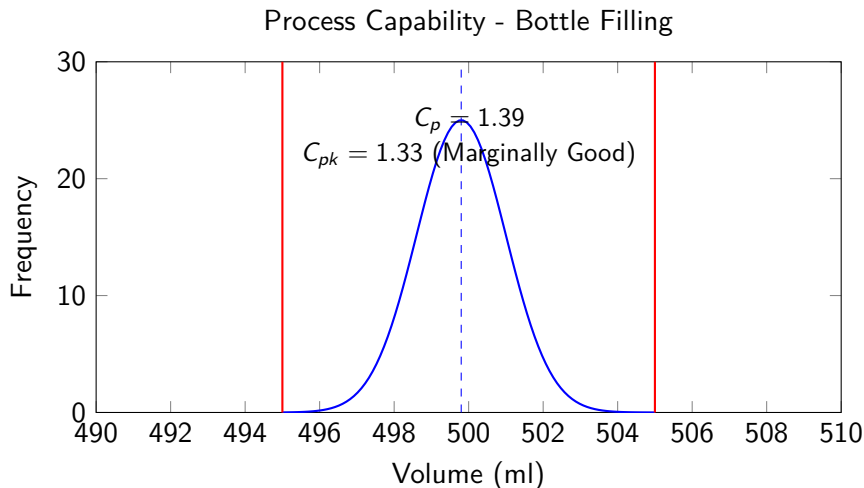
## Process Data:

- Process Mean ( $\mu$ ) = 499.8 ml
- Short-term Standard Deviation ( $\sigma$ ) = 1.2 ml

$$C_p = \frac{505 - 495}{6 \times 1.2} = \frac{10}{7.2} \approx 1.39$$

$$C_{pk} = \min \left( \frac{505 - 499.8}{3 \times 1.2}, \frac{499.8 - 495}{3 \times 1.2} \right) = \min(1.44, 1.33) = 1.33$$

# Bottle Filling – Capability Diagram



# How Capability Curves Are Constructed

- Capability curves are based on the **normal distribution** of process data
- The curve represents the **probability density** of the process output
- Three key elements:
  - ▶ Process Mean ( $\mu$ )
  - ▶ Process Standard Deviation ( $\sigma$ )
  - ▶ Specification Limits (LSL and USL)

# Step-by-Step Construction of Capability Curve

- 1 Collect process data and confirm stability (using Control Charts)
- 2 Calculate process mean ( $\mu$ ) and standard deviation ( $\sigma$ )
- 3 Plot the normal distribution curve using:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- 4 Draw vertical lines for LSL and USL
- 5 Calculate Cp and Cpk values
- 6 Interpret the position and spread of the curve

# Illustration of Example 1: Bottle Filling (Good Process)

## Process Data:

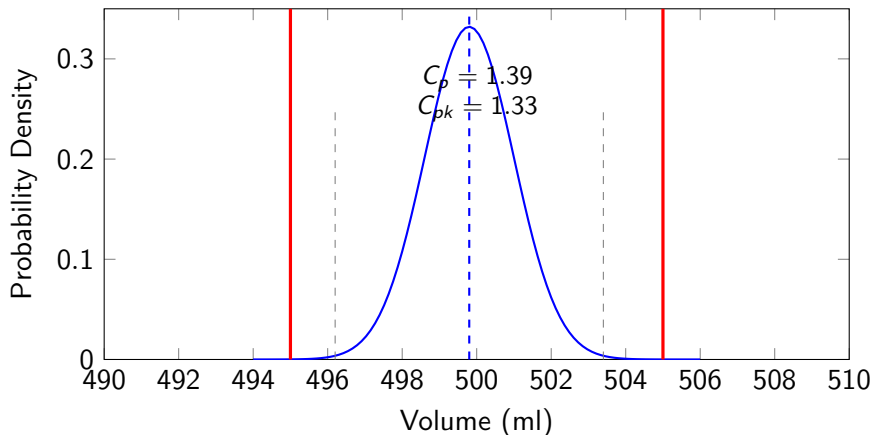
- Target = 500 ml
- LSL = 495 ml, USL = 505 ml
- $\mu = 499.8$  ml
- $\sigma = 1.2$  ml

## Curve Construction:

- Center the normal curve near 499.8 ml
- Spread determined by  $\sigma = 1.2$  (most data within  $\pm 3\sigma = \pm 3.6$  ml)
- Result:  $C_p = 1.39$ ,  $C_{pk} = 1.33$

# Bottle Filling – Curve Construction

Capability Curve Construction - Bottle Filling



## Practical Example 2: Shaft Diameter (Poor Centering)

**Scenario:** Steel shaft diameter LSL = 49.80 mm, USL = 50.20 mm, Target = 50.00 mm

**Results:**

- $\mu = 50.12$  mm (shifted toward upper limit)
- $\sigma = 0.04$  mm

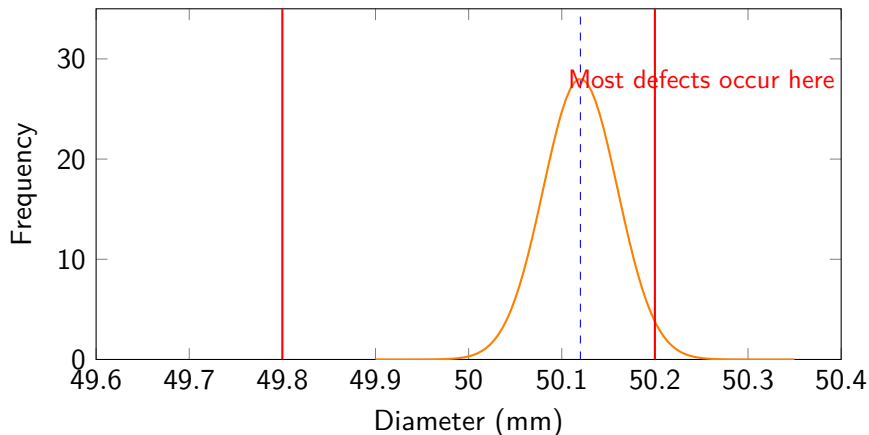
$$C_p = \frac{50.20 - 49.80}{6 \times 0.04} = 1.67 \quad (\text{Good Spread})$$

$$C_{pk} = \min \left( \frac{50.20 - 50.12}{0.12}, \frac{50.12 - 49.80}{0.12} \right) = \min(0.67, 2.67) = 0.67$$

**Conclusion:** Process has good potential but is poorly centered → High defect risk on upper side.

# Shaft Diameter – Off-Center Process

Poor Centering -  $C_{pk} = 0.67$



# Illustration of Example 2: Off-Center Process (Shaft Diameter)

## Process Data:

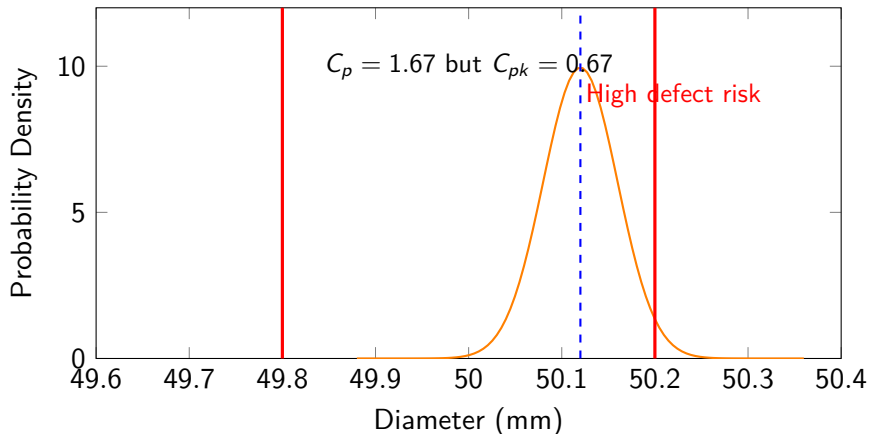
- $LSL = 49.80$  mm,  $USL = 50.20$  mm
- $\mu = 50.12$  mm (shifted)
- $\sigma = 0.04$  mm

## Construction Steps:

- Plot normal curve centered at 50.12 mm
- Spread =  $\pm 3\sigma = \pm 0.12$  mm
- Most output is closer to USL  $\rightarrow$  lower  $C_{pk}$

# Off-Center Process – Curve Construction

Capability Curve - Off-Center Process



## Practical Example 3: Big Data Context - Sensor Temperature

**Scenario:** Temperature control in a chemical reactor (Big Data)

- Specification: 195°C – 205°C
- Data collected: 50,000 readings per day via IoT sensors
- Short-term  $\sigma = 1.8^{\circ}\text{C}$
- Mean = 200.5°C

$$C_p = \frac{10}{6 \times 1.8} \approx 0.93 \quad (\text{Not Capable})$$

$$C_{pk} = 0.83$$

**Insight:** Even with massive data, the process spread is too wide. Needs process improvement.

# Capability Analysis – Decision Making

| Cpk Value     | Status    | Action Required                       |
|---------------|-----------|---------------------------------------|
| $> 1.67$      | Excellent | Maintain                              |
| $1.33 - 1.67$ | Good      | Monitor closely                       |
| $1.00 - 1.33$ | Marginal  | Improve centering or reduce variation |
| $< 1.00$      | Poor      | Major process improvement needed      |

# Key Insights from Curve Construction

- The **width** of the bell curve is determined by  $6\sigma$
- The **position** of the bell curve is determined by  $\mu$
- $C_p$  looks at width only (potential)
- $C_{pk}$  looks at both width and position (actual performance)
- When the curve is centered and narrow  $\rightarrow$  High  $C_p$  and  $C_{pk}$
- When the curve is wide or shifted  $\rightarrow$  Lower capability

# Summary – How to Read Capability Curves

- Tall & narrow curve inside specs = Excellent capability
- Wide curve = High variation → Low  $C_p$
- Shifted curve = Poor centering → Low  $C_{pk}$
- Curve touching or going outside specs = Not capable

**Best Practice:** Always plot the capability histogram with normal curve overlay for visual understanding.

# Key Lessons from Examples

- High  $C_p$  but low  $C_{pk}$  = Process needs **centering**
- Both  $C_p$  and  $C_{pk}$  low = Process needs **variation reduction**
- Stability first → Then Capability Analysis
- In Big Data, we can calculate capability dynamically (rolling windows)
- Always link capability results to business decisions and improvement projects

# Process Capability in Big Data Context

- Analyze millions of data points instead of small samples
- Use rolling capability analysis (moving windows)
- Handle non-normal data with transformations or percentile-based indices
- Real-time capability monitoring using Spark Streaming
- Multivariate capability analysis for correlated parameters

# Key Takeaways

- Process Capability tells us if the process can meet customer specifications
- Stability (via Control Charts) must come before Capability Analysis
- $C_p$  measures spread,  $C_{pk}$  measures spread + centering
- Aim for  $C_{pk} \geq 1.33$  for most processes
- In the Big Data era, capability analysis becomes more accurate and dynamic

# Acceptance Sampling – Introduction

- Statistical method used to decide whether to accept or reject a lot (batch) of products
- Applied to incoming raw materials, finished goods, or outsourced components
- Not a substitute for process control — it is a form of inspection
- Widely used when 100% inspection is too costly or destructive

# Key Concepts in Acceptance Sampling

- **Lot** — A batch of products produced under similar conditions
- **Sample** — Randomly selected items from the lot
- **Acceptance Number (c)** — Maximum number of defectives allowed in sample
- **Producer's Risk ( $\alpha$ )** — Probability of rejecting a good lot (Type I error)
- **Consumer's Risk ( $\beta$ )** — Probability of accepting a bad lot (Type II error)
- **AQL** — Acceptable Quality Limit (producer's quality level)
- **LTPD** — Lot Tolerance Percent Defective (consumer's quality level)

# Types of Acceptance Sampling Plans

- Four main types of sampling plans are commonly used in practice
- Each plan has different procedures, efficiency, and levels of protection
- Choice depends on inspection cost, administrative complexity, and required discrimination

# 1. Single Sampling Plan

**Definition:** The simplest sampling plan. A single sample of fixed size  $n$  is drawn from the lot, and a decision to accept or reject the entire lot is made based on the number of defectives found.

**Decision Rule:**

- If number of defectives  $\leq c$  (acceptance number)  $\rightarrow$  **Accept** the lot
- If number of defectives  $> c \rightarrow$  **Reject** the lot

**Advantages:** Simple, easy to administer, quick decision. **Disadvantages:** Highest inspection cost.

# Double Sampling Plan – Decision Procedure

## Step-by-Step Decision Flow

- ① Take **First Sample** ( $n_1$ )  $\Rightarrow$  Count defectives ( $d_1$ )
- ② Decision after First Sample:
  - ▶  $d_1 \leq c_1 \Rightarrow$  **Accept Lot**
  - ▶  $d_1 \geq r_1 \Rightarrow$  **Reject Lot**
  - ▶  $c_1 < d_1 < r_1 \Rightarrow$  **Take Second Sample**
- ③ If needed: Take **Second Sample** ( $n_2$ )  $\Rightarrow$  Count defectives ( $d_2$ )
- ④ Final Decision (Combined):
  - ▶  $d_1 + d_2 \leq c_2 \Rightarrow$  **Accept Lot**
  - ▶  $d_1 + d_2 > c_2 \Rightarrow$  **Reject Lot**

**Advantages:** Lower average sample number, better discrimination.

**Disadvantages:** More complex administration.

### 3. Multiple Sampling Plan

**Definition:** An extension of double sampling where up to  $k$  successive samples (usually 3 to 7) may be taken before making a final decision.

**Procedure:**

- Take small samples in stages
- At each stage: Accept, Reject, or continue to next sample
- Each stage has its own acceptance and rejection numbers

**Advantages:** Further reduction in average sample size. **Disadvantages:** Complex to manage and administer.

## 4. Sequential Sampling Plan

**Definition:** The most flexible plan. Items are inspected **one by one**, and a decision is made after each inspection until a clear conclusion (accept or reject) is reached.

**Procedure:**

- Inspect one item at a time
- After each item, update cumulative defectives
- Continue until the acceptance or rejection boundary is crossed

**Advantages:** Lowest Average Sample Number (ASN), highly efficient.

**Disadvantages:** Most complex to administer, requires continuous monitoring.

# Comparison of Sampling Plans

| Aspect               | Single     | Double          | Multiple        | Sequential        |
|----------------------|------------|-----------------|-----------------|-------------------|
| Procedure            | One sample | Up to 2 samples | Up to k samples | One-by-one        |
| ASN (Av. Sample No.) | Highest    | Moderate        | Low             | Lowest            |
| Discrimination Power | Good       | Very Good       | Excellent       | Excellent         |
| Complexity           | Very Low   | Moderate        | High            | Very High         |
| Inspection Cost      | High       | Moderate        | Low             | Lowest            |
| Decision Speed       | Fast       | Moderate        | Slower          | Variable          |
| Best For             | Simple ops | Balanced needs  | Cost sensitive  | Expensive testing |

# Key Takeaways

- **Single Sampling:** Simplest but most expensive
- **Double Sampling:** Good balance between cost and performance
- **Multiple Sampling:** Further improves efficiency
- **Sequential Sampling:** Most efficient in terms of sample size, but most complex
- Larger sample sizes or more stages → Steeper OC Curve → Better protection
- The choice depends on inspection cost, administrative capability, and required risk protection

# Operating Characteristic (OC) Curve

The OC Curve shows the **probability of accepting a lot** as a function of the actual quality level (proportion defective).

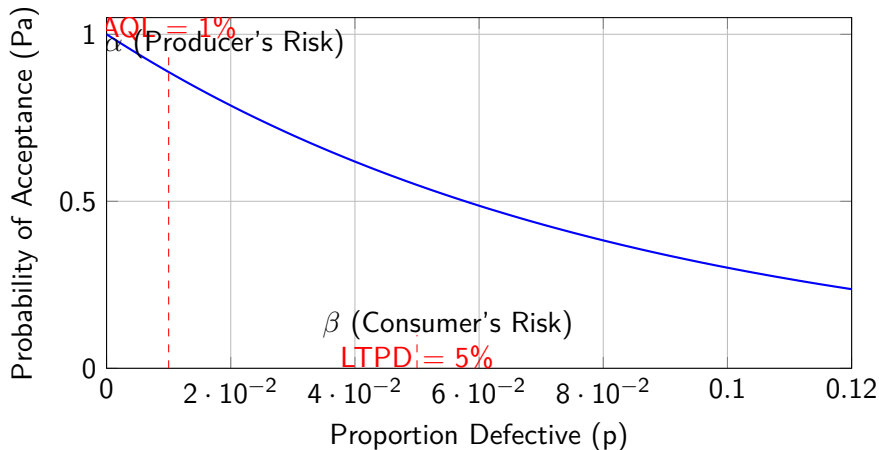
- Ideal OC Curve: Vertical line at AQL
- Real OC Curve: Shows discrimination power of the sampling plan

# Operating Characteristic (OC) Curve in Acceptance Sampling

- The OC Curve is the most important graphical tool in Acceptance Sampling
- It shows the **probability of accepting a lot** as a function of the actual quality level (proportion defective)
- Helps evaluate the performance and risks of a sampling plan

# Key Features of the OC Curve

Typical OC Curve



# Interpretation of the OC Curve

- **Left side (low  $p$ ):** High probability of acceptance  $\rightarrow$  Good quality lots
- **Right side (high  $p$ ):** Low probability of acceptance  $\rightarrow$  Poor quality lots
- **Steepness** of the curve indicates the **discriminating power** of the sampling plan
- Ideal curve would be a vertical line at AQL (perfect discrimination)

# Important Points on the OC Curve

- ① **AQL (Acceptable Quality Limit)**: Quality level the producer aims to deliver. Probability of acceptance is high (usually 95%)
- ② **LTPD (Lot Tolerance Percent Defective)**: Worst quality the consumer is willing to accept. Probability of acceptance is low (usually 10%)
- ③ **Producer's Risk ( $\alpha$ )**: Probability of rejecting a good lot (at AQL)
- ④ **Consumer's Risk ( $\beta$ )**: Probability of accepting a bad lot (at LTPD)

# What the Slope Tells Us

- **Steep OC Curve** = Good sampling plan (strong ability to distinguish good from bad lots)
- **Flat OC Curve** = Poor sampling plan (high chance of wrong decisions)
- **Shifted to the right** = Lenient plan (accepts more lots, higher consumer risk)
- **Shifted to the left** = Strict plan (rejects more lots, higher producer risk)

# Practical Interpretation Example

**Sampling Plan:**  $n = 125$ ,  $c = 3$

- At 1% defective (AQL): Probability of acceptance  $\approx 95\% \rightarrow$  Low Producer's Risk
- At 5% defective (LTPD): Probability of acceptance  $\approx 10\% \rightarrow$  Low Consumer's Risk
- At 3% defective: Probability of acceptance  $\approx 50\%$  (indifference quality level)

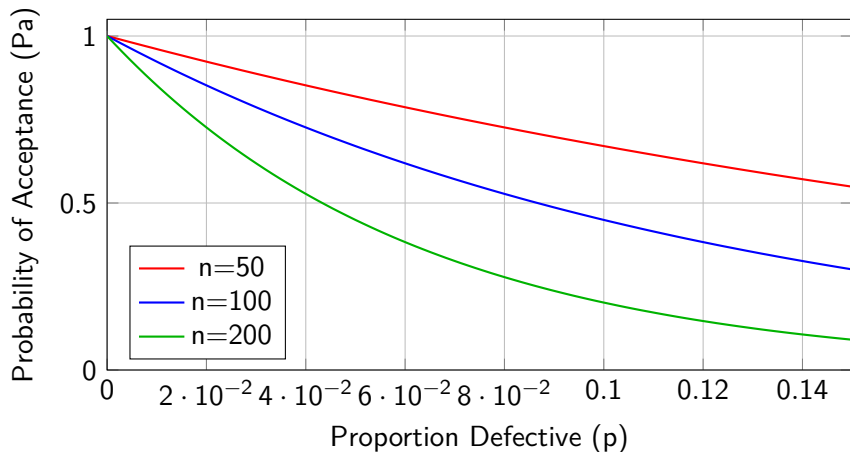
The plan provides reasonable protection to both producer and consumer.

# Comparison of OC Curves for Different Sample Sizes

- The sample size ( $n$ ) significantly affects the performance of a sampling plan
- Larger sample sizes produce **steeper** OC curves
- Steeper curves = Better discrimination between good and bad lots

# OC Curves for Different Sample Sizes ( $c = 2$ )

OC Curves for Different Sample Sizes ( $c=2$ )



# Interpretation of OC Curve Comparison

- **Small sample size ( $n=50$ ):** Flatter curve → Poor discrimination
  - ▶ High chance of accepting bad lots
  - ▶ High Consumer's Risk ( $\beta$ )
- **Medium sample size ( $n=100$ ):** Moderate slope → Reasonable performance
- **Large sample size ( $n=200$ ):** Steep curve → Excellent discrimination
  - ▶ Better protection for both producer and consumer
  - ▶ Lower risks ( $\alpha$  and  $\beta$ )

# Trade-offs with Sample Size

| Sample Size    | Advantages                                 | Disadvantages                                    |
|----------------|--------------------------------------------|--------------------------------------------------|
| Small (n=50)   | Lower inspection cost<br>Faster inspection | Poor discrimination<br>Higher Consumer's Risk    |
| Medium (n=100) | Balanced cost                              | Moderate risks                                   |
| Large (n=200)  | Excellent discrimination<br>Lower risks    | Higher inspection cost<br>Longer inspection time |

# Key Takeaways – Effect of Sample Size on OC Curve

- Increasing sample size makes the OC curve **steeper**
- Steeper curve = Better ability to distinguish good lots from bad lots
- Larger samples provide **better protection** to both producer and consumer
- However, larger samples increase inspection cost and time
- Optimal sample size is a balance between statistical performance and economic considerations

# Key Takeaways – OC Curve Interpretation

- The OC Curve shows the **performance** of a sampling plan
- It helps balance Producer's Risk ( $\alpha$ ) and Consumer's Risk ( $\beta$ )
- Steeper curves = Better discrimination
- Used to design and evaluate sampling plans
- Essential tool for making informed acceptance/rejection decisions

# Advantages and Disadvantages of Acceptance Sampling

## Advantages

- Less expensive than 100% inspection
- Less damage to products
- Useful for destructive testing
- Provides clear decision rules

## Disadvantages

- Risk of accepting bad lots or rejecting good lots
- Does not improve the process
- Should not replace SPC
- Requires careful plan design

# Six Sigma – Introduction

- Data-driven methodology aimed at reducing process variation
- Goal: Achieve \*\*3.4 Defects Per Million Opportunities (DPMO)\*\*
- Originated at Motorola (1986) by Bill Smith
- Popularized by General Electric under Jack Welch
- Combines statistical tools with business strategy

# Six Sigma Performance Levels

| Sigma Level | DPMO    | Process Yield |
|-------------|---------|---------------|
| $6\sigma$   | 3.4     | 99.99966%     |
| $5\sigma$   | 233     | 99.977%       |
| $4\sigma$   | 6,210   | 99.38%        |
| $3\sigma$   | 66,807  | 93.32%        |
| $2\sigma$   | 308,537 | 69.15%        |

# DMAIC – The Six Sigma Roadmap

- 1 **Define** — Define problem, goals, customer requirements (SIPOC, Voice of Customer)
- 2 **Measure** — Collect baseline data, validate measurement system (Gauge R&R)
- 3 **Analyze** — Identify root causes (Fishbone, Pareto, Hypothesis Testing, Regression)
- 4 **Improve** — Develop and implement solutions (DOE, Kaizen, Pilot)
- 5 **Control** — Sustain gains (Control Charts, Control Plans, Dashboards)

# Six Sigma Roles & Certification

- **Champions** — Senior managers who sponsor projects
- **Master Black Belts** — Experts who train and mentor
- **Black Belts** — Full-time project leaders
- **Green Belts** — Part-time project team members
- **Yellow Belts** — Basic awareness

# Lean Six Sigma

- Combines **Six Sigma** (variation reduction) with **Lean** (waste elimination)
- Focus on both quality and speed
- Key Lean Tools: 5S, Value Stream Mapping, Kanban, Poka-Yoke, Just-in-Time
- 8 Wastes (Muda): Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, Unused Talent

# Six Sigma in Big Data Analytics Context

- Improve quality of data pipelines and ETL processes
- Reduce model drift and prediction errors
- Optimize A/B testing and experimentation
- Improve reliability of real-time analytics systems
- Reduce Cost of Poor Data Quality
- Apply DMAIC to machine learning model performance

# Key Takeaways

- Acceptance Sampling is for lot acceptance decisions with known risks
- OC Curve helps evaluate the performance of a sampling plan
- Six Sigma aims for near-perfect quality through variation reduction
- DMAIC is the structured problem-solving framework
- Lean Six Sigma combines speed and quality
- These tools are highly relevant for Big Data quality improvement projects

# Total Quality Management (TQM) – Introduction

- A comprehensive management philosophy focused on continuous improvement of quality
- Involves **everyone** in the organization — from top management to frontline employees
- Shifts focus from product inspection to **process improvement** and **customer satisfaction**
- Foundation for modern frameworks like Six Sigma, Lean, and ISO 9001

# Key Contributors to TQM

| Expert            | Major Contribution                                    |
|-------------------|-------------------------------------------------------|
| W. Edwards Deming | 14 Points for Management, PDCA Cycle, System of       |
| Joseph Juran      | Quality Trilogy (Planning, Control, Improvement), Pa  |
| Philip Crosby     | "Quality is Free", Zero Defects, Cost of Poor Quality |
| Kaoru Ishikawa    | Fishbone Diagram, 7 QC Tools, Company-wide Qual       |
| Genichi Taguchi   | Robust Design, Loss Function                          |

# Core Principles of TQM

- **Customer Focus** — Internal & external customers
- **Total Employee Involvement** — Empowerment and teamwork
- **Process-Centered Approach** — Manage processes, not just outcomes
- **Integrated System** — All departments work toward common goals
- **Strategic & Systematic Approach**
- **Continual Improvement** (Kaizen)
- **Fact-Based Decision Making** — Use data and statistical tools
- **Effective Communication**
- **Leadership Commitment**

# Deming's 14 Points for Management (Summary)

- Create constancy of purpose
- Adopt the new philosophy
- Cease dependence on inspection
- End awarding business on price tag alone
- Improve constantly the system
- Institute training on the job
- Institute leadership
- Drive out fear
- Break down barriers between departments
- Eliminate slogans and targets
- Eliminate numerical quotas
- Remove barriers to pride of workmanship
- Institute education and self-improvement
- Put everybody to work on the transformation

# Cost of Poor Quality (COPQ)

## Four Categories (Crosby & Juran)

- **Prevention Costs** — Training, process planning, quality audits
- **Appraisal Costs** — Inspection, testing, calibration
- **Internal Failure Costs** — Rework, scrap, downtime
- **External Failure Costs** — Warranty, returns, lost goodwill, reputation damage

**Key Message:** Prevention is much cheaper than failure.

# TQM Implementation Framework

- 1 Top Management Commitment
- 2 Customer Focus & Voice of Customer (VoC)
- 3 Employee Training and Empowerment
- 4 Process Management & Standardization
- 5 Continuous Improvement (PDCA / Kaizen)
- 6 Fact-Based Management (SPC, Data Analytics)
- 7 Recognition and Reward Systems
- 8 Supplier Partnership

# TQM in Big Data Analytics Context

- Data Quality Management as core part of TQM
- Quality of algorithms and machine learning models
- Real-time quality monitoring of data pipelines
- Continuous improvement of analytical processes
- Building quality culture in data science teams
- Integration with DevOps (DataOps / MLOps)
- Reducing "Data Debt" and technical debt

# Benefits of TQM

- Higher customer satisfaction and loyalty
- Reduced costs and waste
- Improved process efficiency and productivity
- Better employee morale and engagement
- Enhanced competitive advantage
- Foundation for Industry 4.0 and data-driven organizations

# Big Data Applications in SQC – Why It Matters

- Traditional SQC was designed for small samples and batch processes
- Modern industries generate massive volumes of high-velocity data (IoT, sensors, logs)
- Big Data enables **real-time**, **predictive**, and **automated** quality control
- Combines classical statistical tools with modern data engineering and machine learning
- Creates significant competitive advantage through reduced defects, lower costs, and higher customer satisfaction

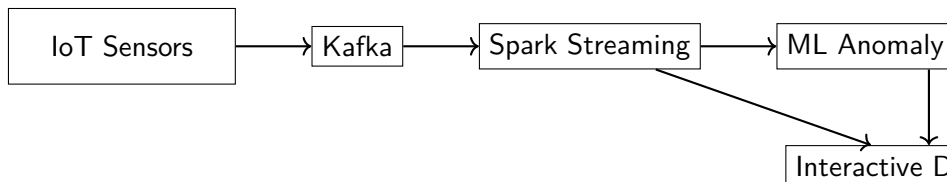
# Traditional SQC vs Big Data-Enabled SQC

| Traditional SQC             | Big Data SQC                                    |
|-----------------------------|-------------------------------------------------|
| Small sample sizes          | Millions of data points per minute              |
| Periodic (batch) analysis   | Real-time streaming analysis                    |
| Manual chart interpretation | Automated anomaly detection                     |
| Reactive                    | Predictive & Prescriptive                       |
| Limited to manufacturing    | Applies to manufacturing, services, software, I |

# Real-Time Statistical Process Control

- Traditional control charts updated every few hours/days
- Big Data enables **\*\*sub-second\*\*** updates
- Technologies: Apache Kafka, Spark Streaming, Flink, Apache Storm
- Dynamic control limits that adapt to data volume and velocity
- Automated alerts and root cause analysis

# Architecture for Real-Time SPC



# Advanced Control Charts for Big Data

- **EWMA** (Exponentially Weighted Moving Average) — Sensitive to small shifts
- **CUSUM** (Cumulative Sum) — Excellent for detecting small persistent shifts
- Adaptive control limits using large historical data
- Multivariate Control Charts (Hotelling's  $T^2$ ) for correlated variables
- Profile Monitoring for functional data (e.g., waveforms)

# Machine Learning + SPC for Anomaly Detection

- Traditional 3-sigma rules + Modern ML algorithms
- Unsupervised techniques:
  - ▶ Isolation Forest
  - ▶ Autoencoders
  - ▶ One-Class SVM
- Hybrid approach: Use control charts for interpretability + ML for detection power
- Real-time scoring on streaming data

# Large-Scale Process Capability Analysis

- Traditional: Small samples, normality assumption
- Big Data:
  - ▶ Non-parametric capability indices
  - ▶ Rolling capability analysis on sliding windows
  - ▶ Handling massive datasets with Spark / Dask
  - ▶ Capability analysis across multiple dimensions (multivariate)
- Predictive Capability using time-series forecasting

# Predictive Quality Analytics

- Move from reactive to **\*\*predictive\*\*** quality
- Predict defects before they occur using:
  - ▶ Time-series forecasting
  - ▶ Survival analysis
  - ▶ Deep Learning on sensor patterns
- Digital Twin + Quality Prediction
- Remaining Useful Life (RUL) prediction for equipment

# Case Study 1: Smart Manufacturing

**Scenario:** Automotive assembly line with 10,000+ sensors

- Real-time  $\bar{X}$  and EWMA charts on Spark
- Anomaly detection using Isolation Forest
- Predictive maintenance reduced unplanned downtime by 35%
- Automated quality alerts sent to operators via mobile

## Case Study 2: Data Pipeline Quality (Big Data ETL)

**Problem:** High error rate in daily ETL pipeline

- Implemented real-time p-chart on data validation results
- Used ML to detect anomalous data patterns
- Reduced error rate from 2.8% to 0.3%
- Built self-healing data quality rules

# Challenges in Big Data SQC

- Volume, Velocity, Variety, Veracity
- High-dimensional data
- Non-stationary processes
- Data quality of the quality data itself
- Interpretability of black-box ML models
- Computational cost and scalability

# Opportunities for Big Data Analysts

- Develop end-to-end quality monitoring platforms
- Build hybrid SPC + ML systems
- Create interactive quality dashboards (Power BI, Grafana, Superset)
- Design automated closed-loop quality control systems
- Research in areas like Deep SPC, Quantum SPC, and Edge Computing quality

# Key Takeaways

- Big Data transforms SQC from periodic inspection to real-time intelligent monitoring
- Integration of classical tools (Control Charts, Capability) with modern technologies is powerful
- Predictive and prescriptive quality is the future
- Data quality itself must be managed as a core process
- MSc Big Data graduates are ideally positioned to lead quality transformation initiatives